

MechanicaFluidorum

A Verifier-in-the-Loop Program Motivated by 3D Navier–Stokes Regularity

Dual-Scale Geometry, the Symmetric-Square Lock, and
the Truncated Viscous Katz–Pavlović Dyadic Shell Model

*Scientific Progress Report — Intermediate Results,
Revised After the External Audit of 13 August 2026*

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SocrateAI-Scientific-MechanicaFluidorum

13 August 2026 | revised 12 September 2026

Repository state: commit `af2c96e` | All verification gates PASS (exit 0)

12 Lean files, 258 kernel-compiled theorems, zero `sorry` | 23 exact-arithmetic harnesses

Abstract

This report documents the state of a staged, verifier-in-the-loop research programme on the global regularity of the three-dimensional incompressible Navier–Stokes equations (NSE), organised around *Hypothesis U* — a uniform-in-cutoff enstrophy bound for the frequency-truncated system. The programme’s distinguishing feature is not a claimed proof but a *method*: every assertion carries an explicit epistemic tier, mechanically enforced by a two-gate verification pipeline (exact rational arithmetic; Lean 4 kernel compilation with a pinned axiom footprint). We report 72 theorems kernel-compiled across seven Lean files — of which 65 carry Tier A claims, the remaining 7 having been *demoted* to Tier C by the audit below — six exact-arithmetic certification harnesses, and, equally importantly, a documented record of what has *failed*: two abandoned numerical designs, five caught errata (one in a formula sourced from the literature, one in the programme’s own design memo, one a broken commit, one a false positive in the verification gate itself, and one an unpinned dependency that would have verified the programme against the wrong prover library), four analytical tracks that remain correctly blocked on undefined objects, and — the largest single failure in the programme’s history — **an external expert audit that killed this report’s own central framing claim (§2). The retraction is stated in full and accepted without qualification**: the formalisation, with a one-dimensional index cutoff, does not restate the unreduced three-dimensional problem; it describes a dyadic shell hierarchy. The programme has been re-targeted accordingly, and its formal object is now the truncated viscous Katz–Pavlović dyadic shell model. **No claim is made that Hypothesis U is true, in any form, nor that the Navier–Stokes problem is resolved.** The Millennium Prize problem remains open and is not addressed by anything proven here.

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1 Context Management: Reading This Document and This Repository

Purpose of this section

This programme is executed collaboratively by a human mathematician and AI agents across many sessions. Context is therefore a *managed artifact*, not an accident of memory. This section specifies where authoritative context lives, how it is kept honest, and what a reader (human or machine) must load before acting. It is placed first deliberately: acting on this programme without this context has already caused documented failures (§14).

1.1 The governing documents and their precedence

Context is deliberately *not* held in any individual's or agent's memory. It is externalised into five documents with a strict precedence order:

Document	Role	Authority
SPEC.md	Normative rulebook	Highest. Defines the tier system, the honesty clause, the obstruction ledger. Where any other document conflicts with it, SPEC.md wins.
PLAN.md	Operational law	Task list with Definition-of-Done criteria and the escalation protocol (E-1...E-5). Wins over calendar/roadmap.
LEDGER.md	Claim inventory	Normative: a claim not listed here has no tier and may not be cited.
LL.md	Lessons learned	Why each process rule exists, with the incident that produced it.
CLAUDE.md	Build/verify manual	Commands, architecture, Mathlib location, working conventions.

1.2 The epistemic charter

The organising principle, stated in SPEC.md §0, is a division of labour:

The machine verifier checks the proofs; the human mathematician audits the questions.

Machine verification establishes that a proof is valid. It cannot establish that a *statement* means what the mathematics intends — that a formalisation is *adequate* to its informal gloss. Adequacy is reserved to human audit. Consequently **no LLM output gates a tier promotion**, and several artifacts in this repository are compiled, kernel-clean, and nonetheless explicitly marked DRAFT pending human statement-adequacy audit (§6).

1.3 The three-tier gating system

Tier	Meaning	Mechanically enforced gate
Tier C	Conjecture, analogy, physical narrative, unverified reduction	None, but must be tagged. Confined to <code>exploration/</code> or <code>docs/narrative/</code> . Floating-point code lives here and <i>only</i> here.
Tier B	Checkable: identities validated in exact rational arithmetic	Harness exits 0. <code>fractions.Fraction</code> or <code>int</code> only — floats are banned. Every harness ships a negative control demonstrated to actually fail.
Tier A	Established: Lean 4 kernel-compiled	Zero <code>sorry</code> ; axiom footprint naming <i>no axiom outside</i> <code>{propext, Classical.choice, Quot.sound}</code> . A strict subset is cleaner and passes — see §14, where the earlier literal-equality reading is recorded as a defect.

Two subtleties are load-bearing and were learned the hard way:

1. **“No sorry in the source” is not the Tier A gate.** A failed proof still defines its theorem name in the environment; only `#print axioms` reveals a stray `sorryAx`. The gate is the *axiom footprint*, checked on the compiled artifact.
2. **A checker that cannot fail is not a checker.** Every Tier B harness must contain a negative control that is *demonstrated* to fail when the perturbation is applied. This report documents cases where an intended negative control turned out to be undetectable by construction, and had to be redesigned (§9).

1.4 The honesty clause and the obstruction ledger

Every public claim carries its tier. Tier A status attaches to the *formal statement actually proven*, never to its informal gloss. Additionally, `SPEC.md` §1.3 requires every proposed strategy — human or agent-generated — to file an *Obstruction Compliance Note* addressing five known barriers (Table 1). Strategies that cannot are rejected at review, before any formal work begins.

ID	Obstruction	Compliance question the strategy must answer
O1	Tao’s averaged NSE blow-up	Which structural feature of the <i>true</i> nonlinearity, beyond the energy identity, does the strategy use?
O2	Supercriticality of enstrophy	Where exactly does the strategy gain over the naive enstrophy inequality, <i>uniformly</i> in α' ?
O3	Caffarelli–Kohn–Nirenberg	If the strategy implies something stronger than CKN, what new mechanism pays for it?
O4	Buckmaster–Vicol non-uniqueness	In which class is the $\alpha' \rightarrow 0$ limit taken, and why is the limit the physical solution?
O5	The Euler test	Point to the step that fails at $\nu = 0$. A strategy that would also “prove” 3D Euler regularity is presumptively wrong.

Table 1: The obstruction ledger (`SPEC.md` §1.3). These are the barriers any credible attack must address; they are stated here because they constrain everything in §7. **Re-scoped 2026-08-13** by audit verdict D2, which found that merely *listing* these barriers against a one-dimensional sequence model overstates contact with them: O1 (Tao) and O5 (the Euler test) remain meaningful for the dyadic target; O2–O4 are inherited only if and when the 3-D bridge is built.

1.5 Reproducing the verification state

The only thing that counts as “it works” is the verification pipeline:

```
./scripts/verify.sh  # Gate 1  : Tier B exact-arithmetic harnesses
                    # Gate 1b : LEDGER consistency + single-active-file lint
                    # Gate 2  : Lean kernel compile + axiom-footprint membership test
```

Every dataset in `data/` ships a `.meta` sidecar recording the exact generating command, tool versions, and `sha256sum`, so results can be audited rather than merely re-read.

Gate 1b (added 2026-08-13). `LEDGER.md` is normative — a claim not listed there may not be cited — so an *orphan* row, one citing a theorem or file that no longer exists, silently degrades the whole inventory. Gate 1b checks that every repo path cited in the ledger exists, that every Lean name in a Tier A row is actually declared, and that no banned versioned duplicates (`_v2`, `_final`) have appeared. It carries its own negative controls, run on every invocation: a dangling path, a dangling name, and a banned filename must each be *rejected*, and a clean case must be *accepted*. It found one real defect on its first run (a row citing the non-identifier fragment `_half` rather than `step1_flux_bound_half`).

A latent reproducibility defect, found by doing the cold build

`lean_src/` was made a standalone Lake project on 2026-08-13, so Gate 2 no longer depends on an external Mathlib checkout this repository does not control. (That coupling was not hypothetical: an unrelated build running in that shared checkout rebuilt its `.olean` files mid-run and failed Gate 2 on a file this repository had not touched.)

Doing the build surfaced something more serious than the inconvenience that motivated it. The project’s `lakefile` declared its dependency as `require mathlib from git with no revision pinned`. A cold build from that specification resolves to Mathlib *master* — which in general does not build against the pinned toolchain, and, worse, would have silently verified every result against a *different Mathlib* than the one all existing `LEDGER.md` claims were checked with. For a programme whose entire epistemic claim rests on machine-checkability, an unpinned verifier is a defect in the foundation, not in the build script. Mathlib is now pinned by revision, and the lockfile pinning every transitive dependency is tracked rather than ignored.

2 The Problem, Stated Honestly

2.1 Classical setting

The incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$ are

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0. \quad (1)$$

Leray established global existence of weak solutions [1]. The Clay Millennium Prize problem, as formalised by Fefferman [2], asks for a proof of *exactly one* of four statements — (A) existence and smoothness on $\mathbb{R}^3$; (B) existence and smoothness on $\mathbb{R}^3/\mathbb{Z}^3$ (the periodic case, which is the setting of this programme); (C) breakdown on $\mathbb{R}^3$; (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. Note that a *breakdown* result resolves the problem just as a regularity result does; this is consistent with the present programme’s stance that a negative verdict is a successful outcome. The obstruction is *supercriticality*: the natural energy bound is too weak to control the nonlinearity at small scales.

2.2 The regularised family and Hypothesis U

For $\alpha' > 0$ let $J_{\sqrt{\alpha'}}$ denote the projection onto Fourier modes $|k| \leq 1/\sqrt{\alpha'}$, and let $u^{(\alpha')}$ solve the truncated system

$$\partial_t u + ((J_{\sqrt{\alpha'}} u) \cdot \nabla) u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = J_{\sqrt{\alpha'}} u_0, \quad (2)$$

which is globally smooth for each fixed $\alpha' > 0$.

Hypothesis 1 (Hypothesis U). For data u_0 and horizon T ,

$$\sup_{\alpha' > 0} \sup_{0 \leq t \leq T} \|\nabla u^{(\alpha')}(t)\|_{L^2(\mathbb{T}^3)} < \infty. \quad (3)$$

Proposition 2.1 (Millennium Reformulation; **Tier C** at paper level). *If Hypothesis U holds for all smooth divergence-free u_0 and all T , then 3D Navier–Stokes is globally regular. The chain is: energy identity $\Rightarrow L_t^\infty L_x^2 \cap L_t^2 H_x^1$ bounds $\Rightarrow$ Aubin–Lions compactness $\Rightarrow$ a strong- L^2 subsequential limit which is Leray–Hopf $\Rightarrow$ under U the limit lies in $L_t^\infty H_x^1 \hookrightarrow L_t^\infty L_x^6 \Rightarrow$ the Serrin class $2/s + 3/q = 1/2 \leq 1 \Rightarrow$ smooth [3].*

Retraction — please read before any other claim in this report

An earlier version of this report carried, in this box, the claim that “Hypothesis U is a restatement of the open core of the Millennium problem, not a reduction of its difficulty”. That claim was submitted to external expert audit on 13 August 2026 and was killed (verdict D1). It is retracted here in full.

The retraction is not that the claim was too strong about *difficulty* — it is that the claim was false about *what had been formalised*. The audit found that the programme’s Lean statements are indexed by a one-dimensional shell index $n \in \mathbb{N}$ rather than a frequency $k \in \mathbb{Z}^3$ (verdict A2: the index flattening destroys the lattice geometry, the density of states, and the triad constraint $k_1 + k_2 = k_3$); that the enstrophy weight was the exponential sequence 4^n rather than the Laplacian symbol $|k|^2$ (A3); and that the mode-interaction term B was an *unconstrained* parameter, so that certifying a bound for it was, in the auditor’s words, “authenticating a mathematically empty envelope” (A5). Taken together, what the formalisation describes is a **dyadic shell hierarchy**, not the unreduced 3-D equations. The prose of (1)–(3) above was never wrong; the claim that the *formal artifacts* restated it was.

The new target. The owner accepted the verdict in full (docs/Memo 1.md) and re-targeted the programme rather than defending the framing. The formal target is now **global regularity bounds for the truncated viscous Katz–Pavlović dyadic shell model** — not a proxy for the 3-D problem, but an object of study in its own right, and a respected active area of mathematical fluid mechanics (blow-up: Katz–Pavlović [7], Cheskidov [8]; regularity under strong dissipation: Barbato–Morandin–Romito [9]). A machine-verified regularity-or-blow-up theorem for that model would, to the authors’ knowledge, be a first. The 3-D bridge — the $\mathbb{Z}^3$ reindexing, OP-2 and OP-6 (§9) — is explicit, open future work, and is never again to be an implicit claim of the formalisation.

Why this is recorded as a success of the method and not merely as a setback. This programme’s Definition of Done (§15) does not include proving Hypothesis U. It includes obtaining human verdicts *whatever they are*. The tier system exists precisely so that a framing claim can be located, isolated, and killed without taking any kernel-verified theorem down with it — and that is what happened: every Lean file still compiles, and the retraction cost exactly the seven theorems that deserved to be demoted (§6). The full verdict table (A1–A5, B1–B5, C1, D1–D2) with dispositions is in LEDGER.md; the retraction text is normative in SPEC.md §1.2.

What survives the retraction, stated precisely

Hypothesis U (3) as an informal statement about the continuum truncated system, and Proposition 5.1 as paper-level mathematics, are unaffected — they were always **Tier C** at paper level and are so labelled above. What is retracted is the assertion that the *Lean formalisation* of §6 is a formalisation of *that* statement. Audit verdicts A1 (the quantifier order $\exists C \forall N$) and A4 (quantifying over every solution) were upheld, so the statement’s *logical shape* was sound; it was its *referent* that was wrong. Under the pivot, A2 and A3 dissolve rather than being repaired: for the dyadic shell model the index *is* the object and the weight $4^n = k_n^2$ *is* the correct weight. A5 was repaired by instantiating B (§6).

3 Dual-Scale Geometry and T-Duality

3.1 Physical motivation (**Tier C**) and its strict boundary

The programme’s geometric inspiration is T-duality in string theory. For a dimension compactified on a circle of radius R , the free-boson partition function is invariant under $R/\sqrt{\alpha'} \rightarrow \sqrt{\alpha'}/R$ with simultaneous exchange of momentum and winding quantum numbers — this is a genuine theorem about the CFT spectrum, established as Eq. (2.2.11) of [14]. The fixed point of that map is $R = \sqrt{\alpha'}$.

Boundary of the analogy — Tier C, quarantined

Reference [14] was retrieved and read for this report, and the distinction it forces is worth stating precisely, because the programme’s earlier internal documents blurred it. The **rigorous** content is the exact $R \leftrightarrow \alpha'/R$ symmetry of the spectrum. The frequently quoted claim that T-duality entails a “**minimal length**” is a *physical interpretation* of that symmetry — the argument being that a string cannot distinguish R from α'/R , so sub- $\sqrt{\alpha'}$ geometry is not resolvable — and is **not itself a separately proved theorem** in the sources consulted.

This report therefore uses T-duality **only** as motivation for the *shape* of a cutoff function. The mechanism acting on the PDE is the Fourier cutoff $J_{\sqrt{\alpha'}}$ and nothing more. Deriving the truncated dynamics from a genuine metric deformation is an *open problem* of this programme (OP-1), not an assumption. All cosmological interpretation — bounce scenarios, dark-sector mass laws, K3/mock-modular numerology — is quarantined in `docs/narrative/`, may never be imported by `lean_src/`, and may never be cited in a tier promotion.

3.2 The effective radius: what is actually proven (**Tier A**)

Define, for $\alpha, R > 0$,

$$R_{\text{eff}}(\alpha, R) := \max\left(R, \frac{\alpha}{R}\right). \quad (4)$$

The following are kernel-verified in `lean_src/CallensDualScale.lean`, with axiom footprint exactly `[propext, Classical.choice, Quot.sound]`:

Formal name	Statement	Reading
<code>Reff_pos</code>	$0 < R_{\text{eff}}(\alpha, R)$	positivity
<code>Reff_ge_sqrt</code>	$\sqrt{\alpha} \leq R_{\text{eff}}(\alpha, R)$	universal minimum scale
<code>Reff_bounce</code>	$R < \sqrt{\alpha} \Rightarrow R_{\text{eff}} = \alpha/R$	sub-cutoff scales reflect
<code>Reff_inertial</code>	$\sqrt{\alpha} \leq R \Rightarrow R_{\text{eff}} = R$	macroscopic scales unmodified
<code>Reff_tdual</code>	$R_{\text{eff}}(\alpha, \alpha/R) = R_{\text{eff}}(\alpha, R)$	self-dual symmetry
<code>Reff_gt_sqrt_of_ne</code>	$R \neq \sqrt{\alpha} \Rightarrow R_{\text{eff}} > \sqrt{\alpha}$	strictness
<code>Reff_eq_sqrt_iff</code>	$R_{\text{eff}} = \sqrt{\alpha} \iff R = \sqrt{\alpha}$	sharpness: the minimum is attained at exactly one point

Scope — what these theorems are and are not

These are theorems about the real-valued function $\max(R, \alpha/R)$. They are elementary, they are correct, and they are *not yet about fluids*. Their role is to fix the geometric language precisely; the parameter α' is carried as a *hypothesis parameter* ($0 < \alpha$), never as an axiom, because Hypothesis U itself quantifies over it. A prior version of this repository asserted these as proven when no such proofs existed; that defect and its correction are recorded in `LEDGER.md` under “Retired / corrected claims”.

3.3 The Symmetric-Square Lock (Tier A)

The programme’s second structural ingredient is an algebraic rigidity constraint. If a sequence satisfies a second-order recurrence with operator L_2 ,

$$u_{n+2} = a u_{n+1} + b u_n,$$

then $v_n := u_n^2$ satisfies the third-order recurrence

$$v_{n+3} = (a^2 + b) v_{n+2} + b(a^2 + b) v_{n+1} - b^3 v_n, \quad (5)$$

which is the symmetric square $L_3 = \text{Sym}^2(L_2)$. If L_2 has characteristic roots $\{\lambda, \mu\}$, then L_3 has roots $\{\lambda^2, \lambda\mu, \mu^2\}$, and the coefficients of (5) are precisely their elementary symmetric functions:

$$e_1 = \lambda^2 + \lambda\mu + \mu^2 = a^2 + b, \quad e_2 = -b(a^2 + b), \quad e_3 = (\lambda\mu)^3 = -b^3.$$

This spectral content is verified as `sym2_symmetric_functions` — it is what makes the identity *structural* rather than a coincidence. The symmetric square of a second-order operator being third-order is classical in the theory of linear differential operators and Picard–Fuchs equations [15].

Intended role (Tier C). The *conjectural* purpose of the lock is to tie macroscopic transport (L_3) rigidly to a microscopic fiber operator (L_2), so that high-frequency modes cannot evolve as independent degrees of freedom — the hoped-for mechanism against Tao-type “choreographed” blow-up [4]. **Its relevance to Navier–Stokes is a conjecture of this programme and is not established.** It must be earned in the dyadic laboratory (§5).

4 The “Holographic Algebra” Organising Concept

Tier and status

The name *HoloAlg* (Holographic Algebra) denotes the programme’s organising picture, not a proven theory. **Everything in this section is Tier C** except where it explicitly cites a Tier A result from §3. It is included because a reader must understand the motivating picture to follow why the four tracks (§7) were chosen — not because the picture carries evidential weight.

4.1 The picture

The organising idea combines two elements already made precise:

1. **A dual-scale geometry** in which the effective scale seen by the fluid is $R_{\text{eff}}(\alpha', R) = \max(R, \alpha'/R)$, so that (i) macroscopic scales are unmodified (`Reff_inertial`: “inertial invisibility”), and (ii) no scale below $\sqrt{\alpha'}$ is ever resolved (`Reff_ge_sqrt`). The regularisation is thereby confined strictly to the sub-cutoff regime.
2. **An algebraic lock** (§3) constraining macroscopic modes to be generated by products of microscopic ones, restricting the admissible spectrum to $\{\lambda^2, \lambda\mu, \mu^2\}$.

The label “holographic” is used in the loose sense that macroscopic (bulk) transport is determined by — and rigidly constrained by — data on a lower-dimensional algebraic structure (the fiber operator’s spectrum). **No claim is made of a relationship to the AdS/CFT correspondence or any holographic duality in the technical sense of theoretical physics.** No such correspondence has been constructed here, and none is used.

4.2 Why this picture would matter, *if* it could be substantiated

For fluid mechanics, the mechanism would matter because it addresses obstruction O1 (Table 1) with a *structural* feature rather than an energetic one: Tao’s averaged system satisfies the energy identity yet blows up, so any successful strategy must use something beyond energy. Algebraic rigidity of the mode spectrum is such a candidate. For astrophysics and cosmology, the same cutoff geometry has been *speculatively* associated with bounce scenarios — but that association is quarantined narrative, carries no evidential weight here, and is deliberately excluded from the mathematical core.

The programme’s own structural finding against itself

When the four tracks’ missing definitions were drafted (`docs/designs/TRACK_DEFINITIONS_DRAFT.md`), the drafting process surfaced a negative structural result about the programme’s own architecture: under the natural choice of entropy functional, **Track T3 collapses into the direct production-identity attack rather than being independent**, and **T1 and T4 both depend on the same unproven embedding (OP-2)**, so agreement between them would be correlated rather than independent evidence. The advertised “four independent attacks” is therefore *not* currently four independent attacks. This finding is recorded rather than suppressed.

5 The Dyadic Shell Model: Established Results

This section was written when the truncated dyadic shell model was the programme’s *laboratory* — a place to earn techniques before touching the continuum PDE. After the audit of 13 August

2026 it is the programme's *target* (§2). The results below are unchanged by that reclassification; what changed is that they are no longer preliminaries to something else. The model is due to Katz and Pavlović [7], whose published form (verified by retrieval, as restated in [8]) is $\frac{d}{dt}u_n + \nu 2^{2\alpha n}u_n - \lambda^n u_{n-1}^2 + \lambda^{n+1}u_n u_{n+1} = g_n$. The system used here is the unforced case with λ^n playing the role of k_n : shells $n = 0, \dots, N$ carry wavenumbers $k_n = 2^n$ and real amplitudes a_n , with

$$\frac{da_n}{dt} = \text{NL}_n - \nu k_n^2 a_n, \quad \text{NL}_n := k_{n-1} a_{n-1}^2 - k_n a_n a_{n+1}, \quad (6)$$

with truncation conventions $a_{-1} = 0$, $a_{N+1} = 0$. Energy is $E = \frac{1}{2} \sum a_n^2$ and enstrophy $\Omega = \frac{1}{2} \sum k_n^2 a_n^2$.

Scope note (honesty clause)

This model is **not** the Navier–Stokes equations. Its nonlinearity is **never** asserted in this repository to be derived from, or to approximate, $(u \cdot \nabla)u$; the audit of 13 August 2026 (§2) exists precisely because an earlier framing let that boundary blur. Results below are exact statements about finite sums and their consequences; no existence theory for the ODE system is invoked. What the model *is* is a well-studied object with a real literature and real open questions — which is what makes it a legitimate target rather than a proxy.

5.1 Energy flux telescoping and monotonicity (Tier A)

In `lean_src/DyadicShells.lean`: the nonlinear energy flux telescopes for *arbitrary* k and a (`dyadic_flux_telescopes`); it vanishes under the truncation $a_{N+1} = 0$ (`dyadic_flux_zero_of_boundary`); and consequently $dE/dt \leq 0$ for $\nu \geq 0$ (`energyRate_nonpos`). Non-vacuity is witnessed: a concrete instance pins $dE/dt = -1$ exactly, so the inequality is not the trivial $0 \leq 0$.

5.2 The exact enstrophy-production identity (Tier A)

The programme's most substantive dyadic result concerns *enstrophy*, the quantity Hypothesis U is about. Unlike energy flux, the enstrophy production sum does *not* telescope for arbitrary k ; it collapses only under the doubling hypothesis $k_{n+1} = 2k_n$, and it collapses not to zero but to a multiple:

Theorem 5.1 (Dyadic enstrophy production; `enstrophy_production_dyadic`). *Under $k_{n+1} = 2k_n$ and truncation $a_{N+1} = 0$,*

$$\sum_{n=0}^N k_n^2 a_n \text{NL}_n = 3 \sum_{n < N} k_n^3 a_n^2 a_{n+1}, \quad (7)$$

so that the enstrophy evolves as

$$\frac{d\Omega}{dt} = 3 \sum_n k_n^3 a_n^2 a_{n+1} - \nu \sum_n k_n^4 a_n^2. \quad (8)$$

This is the dyadic analogue of vortex stretching, made exact: a signed production term competing against dissipation. The coefficient 3 is not fitted — it is $r^2 - 1$ for doubling ratio $r = 2$, and the Tier B harness independently confirms the general formula at $r = 3$ (coefficient 8). A kernel-verified negative control (`negative_control_nondoubling`) establishes that the identity genuinely *fails* for non-doubling k ; the checker can fail.

5.3 A local production bound (Tier A)

Writing $S_N := \sum_{n < N} k_n^3 a_n^2 a_{n+1}$ and $\Omega_N := \frac{1}{2} \sum_{n \leq N} k_n^2 a_n^2$, `lean_src/EnstrophyProductionBound.lean` proves, in a sqrt-free three-step chain (elementary algebra, $\sum x^2 \leq (\sum x)^2$ for nonnegative x , Cauchy–Schwarz):

$$S_N^2 \leq 2\Omega_N^3. \quad (9)$$

Precise scope of (9) — this is where over-reading would be easy

This is a *local, instantaneous, algebraic* bound. It does *not* use dissipation, it does *not* integrate in time, and it does *not* address uniformity in N . Note the exponent: production is bounded by $\Omega^{3/2}$, which is exactly the supercritical scaling that makes the enstrophy differential inequality $\dot{\Omega} \lesssim \Omega^{3/2}$ insufficient for global control. **Inequality (9) is therefore consistent with blow-up and is not evidence for Hypothesis U.** It is infrastructure: an exact, reusable, verified inequality, nothing more. The Tier B mirror further found, honestly, that the negative control breaks Step 1 under non-doubling k but does *not* break the final bound on the tested state family — an unforced negative result, recorded.

6 Statement-Level Formalisation, the Pivot, and the Demotion

6.1 The shell-model Hypothesis U as a formal statement (Tier A, draft)

`lean_src/DyadicShell_Statements.lean` (renamed 2026-08-13 from `HypothesisU_Statements.lean`, Memo Task 1) gives a non-vacuous Galerkin-shape formalisation. Writing $u : \mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R})$ for mode amplitudes, a solution at cutoff N satisfies: (i) $u(0) = \text{truncate } N u_0$; (ii) for every t and every retained mode $n \leq N$, $\frac{d}{ds} u(s, n)|_{s=t} = B_n(u(t)) - \nu w_n u(t, n)$; (iii) $u(t, n) = 0$ for $n > N$. Then

$$\text{HypothesisU} \equiv \exists C > 0. \forall N. \forall u. [u \text{ solves at cutoff } N] \Rightarrow \forall t \in [0, T]. \Omega_N(u, t) \leq C. \quad (10)$$

Why the quantifier order is the entire content

$\exists C \forall N$ — the constant is chosen *before* the cutoff. Swapping to $\forall N \exists C$ would be the trivial statement that each finite-dimensional truncation is bounded on its own, which carries no information about the $\alpha' \rightarrow 0$ limit. That swap is the failure mode this ordering exists to exclude.

Two prior defects were repaired by explicit human decision (recorded as Q1, Q2): initial data must be *projected* per cutoff, else the hypothesis set is empty at low N and the statement holds vacuously; and the weight is fixed concretely to $w_n = (2^n)^2$, so that “enstrophy” is provably a nonnegative sum of squares rather than merely a name. The file also *machine-checks the refutation* of the prior formalisation: quantifying over all smooth fields, without the equation as a constraint, yields a statement that is **provably false** (`unconstrained_bound_false`), since scaling a field by c scales enstrophy by c^2 .

6.2 Closing verdict A5: the nonlinearity is now concrete (Tier A)

Until 13 August 2026 the mode-interaction term B in (10) was an **abstract parameter**, and the previous version of this report named that the decisive open item. The audit put it far more sharply (verdict A5): certifying a bound for an unconstrained B is a “fatal flaw . . . authenticating a mathematically empty envelope” — an arbitrary truncated sequence system simply blows up, so the envelope contains nothing. Under the pivot this was repaired directly, by instantiating B as the Katz–Pavlović shell nonlinearity itself (Memo Task 2):

$$\text{shellB} : B_0(v) = -(k_0 v_0 v_1), \quad B_{m+1}(v) = k_m v_m^2 - k_{m+1} v_{m+1} v_{m+2}. \quad (11)$$

The bottom shell has no influx, so k_{-1} is never evaluated; for $n \geq 1$ this is exactly NL_n of (6). The structural property whose absence the audit called fatal is now a kernel-checked theorem, obtained from an exact telescoping identity:

Theorem 6.1 (Shell energy pairing; `sum_mul_shellB`, `shellB_energy_conservation`). *For all $k, v : \mathbb{N} \rightarrow \mathbb{R}$ and all N , writing $\text{out}_m := k_m v_m^2 v_{m+1}$, each summand satisfies $v_n B_n(v) = \text{out}_{n-1} - \text{out}_n$ (reading $\text{out}_{-1} = 0$), so that*

$$\sum_{n \leq N} v_n B_n(v) = -k_N v_N^2 v_{N+1}, \quad \text{and hence} \quad v_{N+1} = 0 \implies \sum_{n \leq N} v_n B_n(v) = 0.$$

The truncation hypothesis $v_{N+1} = 0$ is clause (iii) of `IsGalerkinSolution`, so this transfers to solutions with no extra assumption: `galerkin_shellB_conservation` states that every Galerkin solution of the concrete model conserves energy exactly, at every time. The viscous term of `IsGalerkinSolution` is $-\nu w_n u_n = -\nu k_n^2 u_n$, which is already the shell model’s dissipation verbatim; together with (11) this makes

$$\text{DyadicShellHypothesisU} := \text{HypothesisU} \nu T (\text{shellB } k) (n \mapsto k_n^2) u_0$$

the programme’s headline statement, in which **nothing abstract remains**. It is a statement about precisely the truncated viscous Katz–Pavlović system — the same model whose exact enstrophy-production identity is proven in §5.

What this does and does not fix

It fixes A5, and only A5. `DyadicShellHypothesisU` is a *statement*; it is not proven, and nothing here is evidence that it is true. The abstract- B scaffolding is retained in the file solely because the refutation theorems and the inhabitation witnesses quantify over it; it carries no headline claim. And instantiating B with the *true 3-D* nonlinearity remains untouched and open — that is OP-6 (§9), now correctly labelled as the 3-D bridge rather than as this file’s missing piece.

6.3 The conditional Millennium reduction (**Tier C**, demoted, draft)

`lean_src/MillenniumReduction.lean` encodes the logical shape of Proposition 5.1 with the two undischarged analytic steps as *named hypothesis parameters*, never axioms:

`millennium_reduction : ( $\forall T$ , HypothesisU)  $\rightarrow$  AubinLionsStatement  $\rightarrow$  ProdiSerrinStatement  $\rightarrow$  GlobalRegu`

The proof is the one-line composition `hPS (hAL hU)`. The intent was that all mathematical weight be parked in the hypotheses’ *types*, so the Lean kernel would track the outstanding debt in the type signature rather than in the axiom environment — the pattern prescribed after a prior version of this work carried a raw `axiom aubin_lions_compactness`.

An adversarial self-review against `SPEC.md`’s actual text found three statement-adequacy gaps, all repaired by explicit decision (Q3–Q5): global regularity had covered only *time*-regularity, and now additionally requires spatial smoothness via membership in every Fourier-side Sobolev class H^s ; the theorem had fixed a single horizon T while concluding a T -independent result, and now quantifies over all T ; and the weight nonnegativity $w \geq 0$ is now an explicit hypothesis.

Demotion to — audit verdict B1

That intent did not survive audit. `AubinLionsStatement` and `ProdiSerrinStatement` are bare `Prop` parameters, and the auditor’s finding (B1) is that bare `Prop → Prop` arrows “completely bypass PDE theory. The Lean kernel is merely verifying a logical tautology ($A \rightarrow B \rightarrow C$)” — which violates the Definition of Done’s requirement of bona fide statements. **The verdict is accepted and the demotion is executed: all 7 theorems in this file are Tier C as of 2026-08-13.** They remain kernel-compiled and wired into Gate 2 so that they cannot silently rot, but they may no longer be cited as Tier A.

The distinction being drawn is sharp and worth stating plainly: parking analytic debt in a hypothesis’s *type* is only honest if that type has content. Here it had none — a `Prop` variable constrains nothing — so the type signature recorded the debt in name only. The repair is Memo Task 4: import or define the required sequence spaces (ℓ^2, ℓ^p) , state the Aubin–Lions and Prodi–Serrin analogues *for the shell model* with their actual topological content, and (audit item B4) specialise the reduction to the concrete object rather than to a generic B . That is the next design task and is **open**; `PLAN.md` §10 records that no implementation may begin before the design memo exists.

One audit item was repairable immediately (B2, revise). A previous independent review had raised, and this report previously recorded as an open candidate issue, whether the untruncated solution may differ per horizon T without a coherence clause. The audit confirmed it: the existential had to be hoisted. It now is — `HasGlobalBoundedLimit` fixes *one* limit trajectory u_{lim} across *all* horizons T , rather than permitting a different limit for each. The Cantor diagonal argument that would actually produce such a trajectory is thereby relocated into `AubinLionsStatement`’s type, where the audit said it belongs, instead of being silently absent; formalising the diagonalisation itself is part of the Task 4 repair.

Audit status of the two files after the pivot

`DyadicShell_Statements.lean` is **Tier A** and DRAFT: it compiles clean with correct axiom footprints, its quantifier shape was upheld (A1, A4), its nonlinearity is now concrete (A5), but it has **not** passed a full human statement-adequacy audit. `MillenniumReduction.lean` is **Tier C** and DRAFT pending the Task 4 repair. The general lesson is the one this section now demonstrates twice over: fixing the gaps found in a self-review (Q3–Q5) did *not* certify that no others remained, and the ones that remained were the structural ones. Authorship — including adversarial self-review — never licenses a claim. Only human audit does, and when it was finally obtained it removed a claim rather than confirming one.

7 The Four Analytical Tracks: Plan and Blocking Status

Stage 3 of the programme proposes to prove Hypothesis U *scale-by-scale* rather than as a monolith, partitioning Fourier space into dyadic shells and controlling $\sum_j E_j^{(\alpha')}(t)$. Four deep analytical disciplines are proposed as independent attacks. Each is an *analogy with a falsification plan*, and each carries a kill criterion agreed in advance.

Status of this section after the pivot

All four tracks are formulated against the *continuum* problem on $\mathbb{T}^3$ and depend on $\mathbb{Z}^3$ lattice structure (T1’s triads, T4’s percolation field). They therefore sit **downstream of the 3-D bridge** (§9), not of the programme’s current formal target. They are retained here unchanged, and unchanged also in their status: blocked. Their blockage was never contingent on the framing that was retracted — it is contingent on four objects that do not exist, listed below.

	Discipline	Proposed mechanism	First falsifiable milestone / kill criterion
T1	Bourgain–Demeter ℓ^2 decoupling [10]	Arithmetic depletion: the Sym^2 constraint starves triadic resonances $k_1 + k_2 = k_3$	Exact count of constrained vs. unconstrained resonant triads. Kill: constrained count grows at the same order $\Rightarrow$ no depletion.
T2	Mouhot–Villani non-linear Landau damping [11]	Phase mixing suppresses “enstrophy echoes” via a Newton-scheme argument in a high-regularity class	A precise <i>definition</i> of an enstrophy echo in the truncated system, plus its exact evolution. Kill: echoes not suppressed even in the dyadic model.
T3	Golse–Saint-Raymond hydrodynamic limits [12]	A relative-entropy functional controls dissipation uniformly in α'	A relative-entropy inequality for the truncated system at fixed α' . Kill: inequality already fails at fixed α' .
T4	Duminil-Copin percolation [13]	High-enstrophy regions form <i>subcritical</i> percolation clusters $\Rightarrow$ singular set of dimension 0	Recover the CKN dimension bound [5] in the lattice formulation. Kill: cannot recover even CKN.

A near-miss worth recording: the correction that would itself have been an error

An earlier draft of this report flagged an apparent defect in the programme’s specification. The spec describes T2 as resting on *Gevrey-class* regularity; a first retrieval pass indicated that Mouhot–Villani’s *core* result is set in *analytic* regularity, and the obvious move was to “fix” the spec by substituting one for the other.

That fix would have been wrong. Re-fetching the abstract directly — rather than relying on the intermediate summary the first pass produced, parts of which were flagged as paraphrase — gives *both*: the paper establishes “exponential Landau damping in **analytic** regularity”, while the version note records that the main result “now covers Coulomb and Newton potentials, and (2) some classes of **Gevrey** data”. The base theorem is analytic; the final version does cover Gevrey. **The spec was defensible as written and was left standing.**

The methodological point is the reason this is in the report at all: the first pass had enough material to justify editing the normative rulebook, and editing it would have *introduced* an error on the strength of a paraphrase — exactly the failure mode rule LL-6 exists to prevent, one level up from where it was first learned. **When a proposed correction rests on a summary rather than a primary quotation, flag it and re-fetch; do not edit the rulebook.**

One substantive caveat did survive, and it is the part that bears on feasibility: coverage is “*some* classes of Gevrey data”, not all. Any T2 argument leaning on Gevrey regularity must identify *which* class and must not assume the full Gevrey scale.

A disanalogy that has not been resolved away

The Golse–Saint-Raymond programme [12] concerns the Boltzmann-to-Navier–Stokes hydrodynamic limit, whereas this programme’s $\alpha' \rightarrow 0$ limit is **not** a Knudsen limit and has no kinetic layer — a disanalogy flagged internally (review finding M4) and repeated here so T3 is not over-read. This one stands.

7.1 Current status: correctly blocked

All four tracks are blocked on undefined objects — and this is the correct state

Tracks T1–T4 require four objects that **do not exist** in this programme or, to the authors’ knowledge, in the literature:

- **OP-2**: the Sym^2 -constrained spectrum on T^3 . The lock is proven for *scalar recurrences*; its embedding into 3D Fourier dynamics indexed by $k \in \mathbb{Z}^3$ is open. Any invented embedding would make T1’s count meaningless.
- **OP-3**: “enstrophy echo” as a formula. Landau-damping echoes are defined in Vlasov phase space; no NSE analogue exists.
- **OP-4**: the entropy functional h and reference state $\bar{u}$. Choosing h is the mathematical content of T3.
- **OP-5**: the coupling of the percolation field to Sym^2 . Without it, T4 measures generic percolation, which cannot beat CKN.

Under escalation rule E-1, an agent encountering a missing definition must **stop and escalate**, never improvise. Draft definitions have been authored and await human audit; *authorship never unblocks a track — only audit does*. Progress here is deliberately gated on human judgement rather than machine throughput.

7.2 Executable instrumentation delivered (Tier B)

Definition-independent tooling has nonetheless been built and certified, so that the tracks can execute immediately once unblocked: exact lattice representation counts $r_3(n)$ for $n \leq 10^4$ with Legendre anchors; exact unconstrained triad counts for $|k_i|^2 \leq M^2$; and an exact 3D periodic site-percolation toolkit (union-find over integers, wrap detection, 27 checks, negative control that fails when periodicity is dropped).

8 Numerical Investigation and Its Two Failures

The central empirical question is whether $\sup_t \Omega_N(t)$ grows with the cutoff N or saturates. Two successive designs *failed their feasibility premises*, for genuinely different reasons; both are documented rather than quietly replaced.

1. **Explicit float RK4 (D4).** Stiffness: stability requires $\Delta t \sim 1/(\nu k_N^2)$, which collapses as N grows. 21 of 36 configurations were computationally infeasible.
2. **Exact-rational IMEX-Euler (D5).** The implicit viscous treatment removed the stiffness barrier as designed — but the scheme failed anyway, for an unrelated reason: iterating a *nonlinear* rational map in exact arithmetic causes the numerators and denominators to grow geometrically in bit-length (hundreds of digits within 2–6 steps), while the state magnitude stays small. **This is a representation blow-up, not a numerical instability.** Implicit-versus-explicit was the wrong axis on which to redesign.

Generalised lesson, now a standing rule

Exact rational arithmetic is excellent for closed-form algebraic identities and inequalities proven in one shot, and poor for *iterating* a nonlinear map over many steps. For trajectories, use dual-precision floating point (Tier C steering) and reserve exactness for the algebra. After two failed designs, the methodology decision was escalated to the human owner rather than a third variant being chosen unilaterally.

8.1 Resolution and the current evidence base (Tier C steering data)

The accepted resolution was to stop pursuing exact-rational certification and retain dual-precision (float64 + 50-digit `mpmath`) runs as permanent Tier C steering data. Across the full grid $N \in \{8, 12, 16, 20, 24\}$, several low-viscosity configurations initially reported divergence, with the two precisions agreeing to $\sim 10^{-14}$ — ruling out rounding, but *not* distinguishing genuine divergence from an under-resolved step size (the design's Δt formula never depended on ν). A step-size refinement study resolved this:

Finding	Detail
Divergences are artifacts	All four diverging (ν , profile) pairs flip to OK at a finite refinement level and stay OK at every finer level, with $\sup \Omega$ <i>decreasing</i> as $\Delta t \rightarrow 0$ (e.g. $45.1 \rightarrow 21.8 \rightarrow 17.3 \rightarrow 15.8$) — the signature of a discretisation artifact resolving, not a fixed-time blow-up.
N -independence	44 of 45 configurations are <i>exactly</i> flat in N (bit-identical $\sup \Omega$ at every cutoff). This was reported as striking evidence. It is not evidence at all — see the retraction immediately below.
The one exception, explained	$\nu = 0.1$, profile P2 appeared to grow linearly in N (4.5, 6.5, $\dots$, 12.5). Traced by hand: this equals $\frac{1}{2}(N + 1)$, which is profile P2's <i>analytic initial</i> enstrophy — P2 excites every retained mode by construction, so a larger cutoff starts with more enstrophy independent of any dynamics. The trajectory collapses immediately from $t = 0$ and decays thereafter. An initial-data artifact, not dynamical growth.

Retraction (14 August 2026): the N -flatness is an artifact of the grid

An earlier version of this report presented the flatness above as the evidence base for a uniformity verdict. **That reading was wrong, and the error was the report's own.** Direct measurement of the shell-amplitude profile shows the cascade never reaches the shells the grid varies. At $\nu = 10^{-3}$ the enstrophy is carried almost entirely by a *single* shell:

shell	peak $ a_n $	$k_n^2 a_n^2$	share of $2\Omega \approx 140$
$n = 8$	4.5×10^{-2}	132.7	$\approx 95\%$
$n = 10$	1.7×10^{-6}	3.0×10^{-6}	2×10^{-8}

The grid's *smallest* cutoff was $N = 8$ — exactly one shell above the decisive one. Raising N from 8 to 24 appended shells contributing $\sim 10^{-8}$ relative, which is why 44 of 45 configurations agreed *bit-identically*: adding numerically zero shells changes $\sup \Omega$ by nothing. **The experiment varied the one parameter along the one range in which it could have no effect.** Lowering to $N = 7$ would have removed 95% of the enstrophy.

“Flat in N ” is therefore close to a tautology here and cannot distinguish “*the bound is uniform in the cutoff*” from “*no regime was tested in which the cutoff binds*”. No verdict may be issued on this evidence base, and none has been.

The failure mode is now mechanically excluded

A Tier B harness (`tests/tier_b_grid_adequacy.py`, wired into Gate 1) rejects such a grid *before it runs*, on exact integer arithmetic. Balancing viscous against nonlinear rates with the K41 amplitude $a_n \sim k_n^{-1/3}$ gives $k_d \sim \nu^{-3/4}$; with $k_n = 2^n$ the cutoff lies at or above the dissipation scale iff $2^{4N}\nu^3 \geq 1$, i.e. writing $\nu = p/q$, iff the integer inequality $2^{4N}p^3 \geq q^3$ holds. No floats, no logarithms.

Its negative control is the historical grid itself: **0 of 15 configurations had a biting cutoff.** The redesigned measurement (`docs/designs/D6_CUTOFF_BITING_MEASUREMENT.md`) sweeps N *across* the dissipation shell instead of above it, reports the production flux at the cutoff and the shell-population profile alongside $\sup \Omega$, and carries a $\nu = 0$ control where blow-up is a published theorem — the sharper control, because it is the one this programme's hoped-for conclusion would predict wrongly.

9 The 3-D Bridge: Instantiating the True Nonlinearity (OP-6)

This section describes work aimed at the *true* 3-D nonlinearity. It should now be read as work on the **bridge** between the programme’s formal target (the shell model, §5, §6) and its motivating problem (§2) — open future work, and not a claim that the bridge exists. Note the distinction from §6: B in (10) is no longer abstract, having been instantiated by the shell nonlinearity (11); what remains open is instantiating it by the Fourier–Galerkin nonlinearity on $\mathbb{Z}^3$, which is a different and strictly harder object.

A repository-wide search established that **no Fourier-triad convolution formula exists anywhere in this programme’s sanctioned content** — the triad tooling counts lattice points without amplitudes; the OP-2 draft self-admits it discards the angular structure of k “which is precisely where 3-D vortex-stretching geometry lives”; and the dyadic NL_n is never asserted to derive from $(u \cdot \nabla)u$. Instantiating B by the 3-D nonlinearity is therefore a genuine E-1 blocker.

On $\mathbb{T}^3$, with $u = \sum_k \hat{u}_k e^{ik \cdot x}$, $k \cdot \hat{u}_k = 0$, $\hat{u}_{-k} = \overline{\hat{u}_k}$, and Leray projector $P(k) = I - \frac{k \otimes k}{|k|^2}$, the truncated nonlinearity is

$$\partial_t \hat{u}_k = -\nu |k|^2 \hat{u}_k - i \sum_{\substack{p+q=k \\ p,q \in \Lambda}} P(k) [(q \cdot \hat{u}_p) \hat{u}_q]. \quad (12)$$

An erratum caught by exact arithmetic — a case study in why the method exists

The formula first recorded in the scoping memo, taken from a single literature cross-check, read $P(k)[q(\hat{u}_p \cdot \hat{u}_q)]$ — the *vector* q times the scalar velocity–velocity dot product. The Tier B harness then *proved*, in exact arithmetic, that this expression is **identically zero for every k** : $P(k)$ annihilates anything parallel to k , and since every pair (p, q) with $p + q = k$ appears alongside its swap (q, p) , the two terms cancel exactly, $P(k)[qs] + P(k)[ps] = P(k)[(p + q)s] = P(k)[ks] = 0$. That cannot be the real nonlinearity. The corrected form (12) — q dotted with $\hat{u}_p$, a scalar, times the vector $\hat{u}_q$ — matches the Fourier transform of the convective derivative and is nonzero. The error was caught by computation, not by inspection.

The resulting harness (`tests/tier_b_nse_triad_convolution.py`, **Tier B**) certifies two structural facts in exact Gaussian-rational arithmetic at $M = 1, 2, 3$: *transversality* $k \cdot N(u)_k = 0$ (unconditional, since $P(k)$ projects onto $k^\perp$), and *detailed energy conservation* $\sum_k \overline{\hat{u}_k} \cdot N(u)_k = 0$ given divergence-free, conjugate-symmetric input. Two negative controls — dropping the Leray projection, and breaking the divergence-free condition on one mode — each break exactly one fact and leave the other intact, matching the hand derivation rather than merely “changing something”.

Recorded honestly: several negative controls were undetectable by construction

Two natural “plausible bug” perturbations turned out to be invisible: swapping $p \leftrightarrow q$ throughout (the pair set is symmetric under relabelling for *every* fixed k , not merely after summing), and substituting the outer k for the summed q (the difference vanishes because $p \cdot \hat{u}_p = 0$ by divergence-freeness). The negative control actually used — multiplying by the wrong velocity factor — was found by *testing* candidates, not by assuming one would work. Additionally, the general triad identity underlying energy conservation is here verified *computationally* at three truncations, **not proven symbolically**; a Tier A proof remains future work.

9.1 From computational check to kernel proof (Tier A)

The harness above certifies two facts *computationally*, at three truncations, and its own docstring declines to claim either as proven. Both are now Tier A, in `lean_src/AbstractAlgebraicConservation.lean` — renamed 2026-08-13 from `TriadConservation.lean` under audit verdict C1, which found the old name OVERSTATES REACH: the file’s abstraction bypasses geometric aliasing on $\mathbb{Z}^3$ entirely, and the name should say so. The docstring now states that exclusion explicitly:

Theorem 9.1 (Termwise triad cancellation; `triad_pairing`). *If $k_p + k_q + k_r = 0$ componentwise and the field is divergence-free at p ($k_p \cdot u_p = 0$), then the two orderings of the triad summand cancel exactly:*

$$(k_q \cdot u_p)(u_q \cdot u_r) + (k_r \cdot u_p)(u_r \cdot u_q) = 0.$$

Why the first attempt failed, and what actually worked

The harness records that a direct *three-way* relabelling (cycling $p \rightarrow q \rightarrow r$) did not close. It does not need to. The cancellation is **two-way**, and stronger than a summation argument: it is **termwise**. Pairing (p, q, r) with (p, r, q) — swapping only the last two indices, leaving p fixed — and using symmetry of the pairing followed by additivity,

$$[(k_q \cdot u_p) + (k_r \cdot u_p)](u_q \cdot u_r) = ((k_q + k_r) \cdot u_p)(u_q \cdot u_r) = (-k_p \cdot u_p)(u_q \cdot u_r) = 0.$$

This was re-derived by hand and confirmed on 6486 triples in exact arithmetic *before* any Lean was written. Summing it over any finite index set closed under the swap gives detailed energy conservation (`triad_sum_zero`); transversality (`transversality_of_sum`) is separate and unconditional, following from the projector landing in $k^\perp$.

Scope: this does not close OP-6

The file is stated over an *arbitrary commutative ring*, with an arbitrary additive index group and additive wavevector map. It is deliberately **not** stated over $\Lambda \subset \mathbb{Z}^3$ with complex velocities, because that concrete apparatus *is* the open decision D1. **The bridge from this lemma to the concrete Fourier–Galerkin setting is not built**; nothing here instantiates B ; nothing here is a statement about Navier–Stokes solutions. One hypothesis is load-bearing rather than technical: the swap has fixed points (where $q = r$, forcing $p = -2q$) at which the pairing yields only $2f = 0$, so the ring must have no 2-torsion. For the intended instantiations $(\mathbb{R}, \mathbb{C})$ this holds.

Open decisions. Instantiating B is not a substitution but a restructuring: the state representation must move from $\mathbb{N} \rightarrow \mathbb{R}$ shell scalars to $\mathbb{Z}^3 \rightarrow \mathbb{C}^3$ Fourier vectors carrying divergence-free and reality constraints. Whether to undertake that full reindexing or seek a smaller shell-preserving proxy, and whether to sequence it before or after OP-2, are open questions reserved to the human owner.

A second, sharper limit on the pivot (14 August 2026)

The audit established that the formalisation describes a shell model rather than 3-D Navier–Stokes. A further fact bounds what the shell model can deliver even in principle.

The dyadic dissipation term is $\nu k_n^{2\alpha}$; the Navier–Stokes viscous term is $\nu \Delta u$, whose Fourier symbol is $-\nu|k|^2$. Hence $\alpha = 1$ is the Navier–Stokes dissipation — the same exponent, not an analogy. Cheskidov proves global regularity for $\alpha \geq 1/2$. Therefore:

The dyadic shell model at Navier–Stokes dissipation is provably regular.

The model’s difficult regime, $\alpha < 1/2$, corresponds to dissipation *weaker* than Navier–Stokes. A result below $\alpha = 1$ is a result about *hypo-dissipative* shell models. That is live mathematics — Cheskidov singles out $\alpha = 1/3$ as carrying the same nonlinear estimates as 4-D Navier–Stokes — but it is **not a step toward 3-D Navier–Stokes and must never be reported as one.**

Correction (E-3b, 15 August 2026): the band is narrower, and there are two of them

Source verification commissioned before opening this track found this report’s own statement of the band to be wrong. **Barbato–Morandin–Romito** (*Nonlinearity* **24** (2011) 3083–3097, Thm A) closed $\alpha \in [2/5, 1/2)$ in 2011 — with *uniqueness* and smoothness, from mere ℓ^2 data — for **positive** initial data. And *both* bounding theorems assume positivity (the blow-up theorem additionally needs *large* data), so there are two bands, not one: for positive data the open region is $[1/3, 2/5)$; for sign-changing data *nothing* is proven below $1/2$ in either direction. With intermittency dimension $d = 5 - 2/\alpha$, the residual positive-data band sits at $d \in [-1, 0)$, outside the physically relevant range — and the field’s own survey (Cheskidov–Dai–Friedlander) states verbatim that BMR “settles that solutions to the dyadic model corresponding to the 3D NSE are globally regular”. **The genuine room is sign-changing data.** The failure mode is worth more than the correction. The Tier B gate encoding these thresholds had *both* controls, and both fired — but its positive-control anchor asserted that $\alpha = 2/5$ is open, which is precisely BMR’s endpoint. A control tests the code against the premise you believe; it cannot test the belief. And LL-6 had already been applied to this very paper — to its *abstract*, which does not carry the β range. New rule (LL-16): a numerical threshold taken from the literature must be sourced to the *theorem statement* with its hypotheses, and a correction ships with a regression control pinning the value that was wrong.

An unplanned consistency check fell out of the same calculation. The K41 balance $\nu k^{2\alpha} \sim k a_k$ with $a_k \sim k^{-1/3}$ gives a dissipation scale $k_d \sim \nu^{-1/(2\alpha-2/3)}$, whose exponent *diverges* as $\alpha \rightarrow 1/3^+$: below that threshold no dissipation scale exists, so the cascade is stopped at no finite shell. That is exactly Cheskidov’s proven blow-up threshold, reached from an elementary heuristic that knows nothing of his proof.

Owner decision: this gap is to be treated as a gap to *close*, not a ceiling to accept. The programme will pursue the bridge — the $\mathbb{Z}^3$ embedding and the instantiation of the nonlinearity — both of which remain blocked on definitions that do not exist, and so admit only authored candidates submitted for human audit.

9.2 Repairing the reduction: giving the hypotheses content (Tier C)

Audit verdict B1 found the reduction’s two hypothesis parameters to be bare $\text{Prop} \rightarrow \text{Prop}$ arrows: “the Lean kernel is merely verifying a logical tautology”. The repair, carried out on 14 August 2026, is *not* to prove those steps — parking undischarged analysis in a named hypothesis parameter rather than an axiom is the pattern this programme deliberately adopted, so that the kernel tracks the debt in the type. The repair is to make the types say something.

What a compactness statement has to deliver

`IsGalerkinLimit` now carries four clauses, each load-bearing: the limit solves the untruncated system; it has *finite enstrophy* (a genuine ℓ^2 condition, since there is no cutoff to truncate the sum); that enstrophy is bounded on every horizon, with the constant quantified *inside* the horizon — compactness gives no uniformity in T , and writing $\exists C \forall T$ would claim more than the analysis provides; and the limit is genuinely the modewise limit of the Galerkin family **along a subsequence**. The subsequence is not a technicality: a compactness argument extracts one, and asserting convergence of the full sequence would overstate the mathematics.

The ℓ^2 upgrade was deliberately *localised*. Hypothesis U quantifies over *Galerkin* solutions, whose truncation clause forces $u_n = 0$ for $n > N$ — so the finite sum over $n \leq N$ already captures every nonzero mode and is *exact* there, not an approximation. Only the objects with no cutoff needed the infinite-series treatment. A bridge lemma (bounded partial sums of nonnegative terms $\iff$ summable with $\sum' \leq C$) makes the upgrade a conservative extension: nothing previously proven is invalidated.

The file remains Tier C, and two things remain unmachine-checkable

The repair makes the undischarged debt *legible*; it does not pay it. Both hypothesis parameters are still assumed. Moreover, whether the four clauses *say the right thing* is a statement-adequacy question reserved to human audit — it cannot be settled by the kernel, and that cost is scheduled rather than discovered later.

9.3 Eliminating a duplication hazard at the root

Until this point every file in `lean_src/` re-declared, verbatim, the definitions it needed from its neighbours, because the verification gate compiled each file in isolation and cross-file imports did not resolve. That convention carried a silent-drift hazard of exactly the kind this programme exists to eliminate: **a correction applied to one copy and not the other leaves both files compiling cleanly, both gates green, and the mathematics divergent**, with no mechanism able to see it.

With the project now self-contained, the gate builds the dependency graph first and then re-elaborates each file with the project’s own build directory on the search path. Imports resolve; the duplication is gone; there is exactly one definition of each object in the repository. One implementation detail is worth recording because getting it wrong would be invisible: the gate must re-elaborate with `lean` rather than rely on `lake build` alone, because a *cached* build target emits no axiom output at all — which would make the footprint check silently vacuous rather than failing loudly.

10 Three Proposed Mechanisms, Killed by Measurement

On 14 August 2026 the programme’s central conjectural mechanism — the Symmetric-Square lock — was tested in three successive formulations. All three failed, each at a cost of hours rather

than months, and each for a variant of the same reason.

Proposed mechanism	What it actually constrains	How it died
Pointwise, $a_{2n} = c a_n^2$	the <i>amplitude</i> ; the spectral slope stays free	On paper: for any power law $a_n = Ar^n$ the ratio is $1/A$, independent of r . Blow-up is characterised by the slope.
Recurrence signature, $c_2^3 + c_1^3 c_3 = 0$	genuinely discriminating (controls separate by 10^{14}) — but nothing specific to blow-up	By measurement: S rises from early to late in the blow-up regimes, <i>and just as much at $\alpha = 1$ where blow-up is provably impossible</i> .
Radial embedding on $\mathbb{Z}^3$, shell indices $\{2i, i+j, 2j\}$	the <i>parity</i> of the shell index	By counting, against a criterion fixed in advance: the permitted fraction <i>rises</i> with M ($0.019 \rightarrow 0.263 \rightarrow 0.525$), and at $M = 8$, 99% of permitted triads are same-shell.

The common defect, and why it is worth naming

Each formulation carried the *symbols* of the proven lock into a new setting while leaving free the quantity it was supposed to control. A translated constraint can look faithful, type-check, pass every gate, and constrain almost nothing. The cheap general test — now a standing rule — is to **compute the constraint’s solution set in the new setting and check it is smaller than intended**: substitute the simplest family the setting supports and see whether the constraint restricts it.

The third death is the most instructive, because the criterion it failed (“*constrained count grows at the same order $\Rightarrow$ no depletion $\Rightarrow$ kill*”) had been written into the specification *before* the mechanism was proposed. The draft’s own honesty clause had even anticipated the cause: a radial reduction discards the angular structure of k , which is where three-dimensional vortex stretching lives. Writing the kill criterion in advance is what turned a disappointment into a result.

What was built in their place

The resonant-triad hypergraph — vertices are Fourier modes, hyperedges are the resonances $k_1 + k_2 = k_3$ — is *definition-independent*: it is fixed by the Navier–Stokes non-linearity itself and needs no blocked definition, so it could be built immediately. Its normalised-Laplacian spectral gap is stable at ≈ 0.8334 across $M = 2, 3, 4$, apparently approaching $\approx 5/6$ (suggestive on three points, not established).

The reading is a constraint on all future proposals: the triad structure is a **strong expander**, so no depletion is available for free, and any candidate lock must be shown to *destroy* that expansion. The vague question “does the locked sub-hypergraph have a spectral gap?” now has a concrete number to beat.

11 The Fourth Translation: Planar Confinement, and the Torus Theorem

The post-mortem of the three deaths of §10 produced a diagnosis rather than a fourth guess: all three translations searched for a *scarcity* mechanism (fewer resonances, smaller exponents) while the proven lock is a *closure* statement. Squaring a velocity field is the Sym^2 operation on its spectrum, $\text{Sym}^2\{k_1, k_2\} = \{2k_1, k_1 + k_2, 2k_2\}$, and the closure of a mode pair under repeated

triadic interaction stays in the 2-plane $\langle k_1, k_2 \rangle$ — computed exactly in this repository: six rounds, 2144 modes generated, zero outside the plane.

OP-2': the lock's exact 3-D content is planar confinement

A flow whose spectrum lies in a 2-plane of $\mathbb{Z}^3$ is a **2D3C** flow (two-dimensional, three-component — the standard term; “2.5-D” also names a different class in parts of the literature). The 3-D equations then split into 2-D Navier–Stokes plus a passively advected scalar (Biferale–Buzicotti–Linkmann, arXiv:1706.02371, quoted verbatim in the design memo), and 2-D regularity is classical (Ladyzhenskaya; attribution to Leray for global strong existence, per Ponce–Racke–Sideris–Titi). **K1, verified in exact arithmetic:** a field supported on the *tilted* plane $\langle (1, 0, 0), (0, 1, 2) \rangle$ has $N(u) = 0$ on all 322 out-of-plane modes under the Leray-projected truncated nonlinearity; the negative control (a generic field) leaks on all 322. The lattice/Galerkin form of this invariance was not found published as such; the continuum antecedents are Moffatt (J. Fluid Mech. 741 (2014) R3, cited by Biferale et al., itself unfetched) and Biferale et al. 2017. This candidate, unlike its three dead predecessors, was screened *before* authoring: the LL-11 vacuity test passes (plane sublattices are a measure-zero subfamily), and the depletion screen scores planes at $D = 1.87 > 1$ — the mechanism was never about triad counts.

11.1 The attractivity experiment (σ), and a trap caught by its null model

The honest research question OP-2' produces is dynamical: *is the planar-locked manifold attractive, neutral, or repulsive under the truncated 3-D Galerkin flow?* The expected answer — repulsive, by the classical instability of 2-D flows in 3-D — is pre-registered as the null, with a theorem-backed positive control (exactly planar data must stay exactly planar; measured at literal 0.00 out-of-plane energy) and a negative control (generic data must show $O(1)$ out-of-plane fraction; measured 0.62–0.74). The pilot measured $\sigma = +0.57$ to $+1.04$: repulsive, ε -independent, faster at lower viscosity.

The full experiment ($M = 3$, coordinate *and* tilted planes, exact rational grid-adequacy stamps) produced one line that exercised the programme's discipline: at $\nu = 1/2$ the measured σ was *negative* (-0.265) — the pre-registered “candidate discovery” branch. Before interpretation, the obvious null model was computed in closed form: under pure linear decay $u_k(t) = u_k(0)e^{-\nu|k|^2 t}$, the out-of-plane fraction *already* drifts at $\sigma_{\text{lin}} = -0.305$, because the out-of-plane modes carry a higher energy-weighted $\langle |k|^2 \rangle$ than the in-plane modes (6.26 vs 4.95). The measured excess $\sigma - \sigma_{\text{lin}} = +0.04 > 0$: the nonlinearity pushes *out* of the plane even there, and the negative sign was spectral bookkeeping. The most seductive artifact is the one that would have announced a discovery; it was caught by the same rule (LL-14/LL-15: every measurement needs the null model computed *before* the number is allowed to mean anything) that caught its predecessor, which had confirmed a hypothesis instead.

The completed table: artifact-corrected excess $\sigma - \sigma_{\text{lin}}$ is **positive at all six measurement points** — $+0.040/ +1.622/ +2.676$ on $z = 0$ and $+0.007/ +0.750/ +2.073$ on the tilted plane at $\nu = 1/2, 1/10, 1/50$ (the $+0.007$ honestly at noise level), ε -independent, with the exact grid-adequacy stamp passing only at $\nu = 1/2$. The positive control held at literal 0.00 out-of-plane energy *on the tilted plane as well* — the K1 invariance theorem exercised dynamically. The pre-registered kill criterion K3 ($\sigma > 0$ everywhere $\Rightarrow$ the mechanism is dead as a route to regularity for *generic* data) is met at every point.

Verdict (owner-issued, 15 August 2026): OP-2' is killed for generic data

The fourth translation of the Symmetric-Square lock is dead as a route to global regularity for generic data — and unlike its three predecessors it died *with a number* and with its mechanism understood: the planar manifold is real, exactly invariant, and **repulsive**, increasingly so as viscosity falls. What the verdict does *not* touch: the geometric half (Sym² closure $\Rightarrow$ exact 2D3C confinement, K1-verified in exact arithmetic including tilted planes, apparently unpublished in its lattice form), the torus theorem, and the 5/6 conjecture — all independent results that merely arose during the screening. Four mechanisms have now been proposed and four killed; the programme's rate of *disproof* is its main output to date, and that is what a falsifiable structure is supposed to produce.

11.2 The neighbourhood theorem, scoped by a full read

A first-pass source check reported that the 2D3C manifold has a published global-regularity neighbourhood (Ponce–Racke–Sideris–Titi, Comm. Math. Phys. **159** (1994) 329–341). A full read of the fetched PDF *narrowed* this claim, and the correction is recorded with the same prominence as the claim: the smallness threshold is Gronwall-exponential ($\delta \lesssim \exp\{-C \int_0^\infty \|\nabla v\|^4\}$, eq. (2.17), constants never tracked in ν), hence not quantitatively exploitable against measured σ amplitudes; the 2-D application (their Theorem 4) is pure two-component 2-D on $\mathbb{R}^2$, unforced, with a decay hypothesis periodic solutions violate — *not* 2D3C and *not* the torus; and no periodic- T^3 case is treated. What survives is the concept — a regularity neighbourhood around near-2-D data in adjacent settings, coexisting cleanly with $\sigma > 0$ — not a citable safety net for the Galerkin torus.

11.3 The torus theorem (Tier A) and the 5/6 ball conjecture

While screening OP-2', the unconstrained resonant-triad 2-section was solved exactly on the torus, and this became the programme's first **Tier A** result about the genuine $\mathbb{Z}^3$ -type resonance structure (**TriadTorus.lean**, 7 theorems, clean footprints; derivation memo written and hand-verified first, per LL-5). For any finite additive abelian group G , $\Lambda = G \setminus \{0\}$, counting (triad, slot-pair) *incidences* — the choice that absorbs the $(u, u, 2u)$ degeneracy with no genericity hypothesis:

$$A(u, v) = 6 - 2 \cdot [u + v = 0] \quad \text{i.e.} \quad A = 6(J - I) - 2P,$$

with P the mode-reversal involution. Spectral corollary (as kernel-verified sum identities): degree $6n - 8$; zero-sum even vectors are eigenvectors with eigenvalue -8 , odd vectors -4 ; normalised spectrum $\{1, -4/(6n - 8), -8/(6n - 8)\}$, so **the torus spectral gap tends to 1**. The measured ball-truncation gap $\approx 5/6$ is therefore a *pure boundary invariant* of the sphere cutoff — an explicitly separated, still-open conjecture (second eigenvalue $1/6$ of the kernel $2 \cdot \mathbf{1}_B(x + y) + 4 \cdot \mathbf{1}_B(x - y)$ on the unit ball). The measured deviations from $5/6$ fall monotonically: 1.7×10^{-3} , 2.4×10^{-4} , 8.6×10^{-5} , 5.9×10^{-5} at $M = 2.5$, 2.5×10^{-5} at $M = 6$ (924 modes, 398,190 triads), and 1.5×10^{-5} at $M = 7$ (1418 modes, 939,930 triads) — seven points, strictly monotone, still **Tier C** evidence for a conjecture, not a result.

12 The Live Band: Formalising Why $\alpha = 1/2$, and What Lies Below

The owner's direction of 15 August 2026 was to target $\alpha \geq 1/2$ as a known theorem — to validate and stabilise — but to *generalise while doing it*, on the view that the real problem is the sign-changing band below $1/2$. Carrying α symbolically through Cheskidov's chain instead of fixing the constant delivers exactly that, and the result is sharper than a formalisation of a known proof would have been.

The threshold is an integrability threshold ([Theorem A](#), `DyadicRiccati.lean`)

Write $y = \|u\|^2$. The bilinear estimate $|(B(u, u), Au)| \leq c_b |Au|^{1/\alpha-1} \|u\|^{4-1/\alpha}$ (source; declared sharp there) admits Young absorption against $-\nu|Au|^2$ iff $\alpha > 1/3$ — which is the content of the *local* theorem, and **not** where $1/2$ comes from. Absorption leaves a Riccati inequality $y' \leq Cy^s$ with $s = (4\alpha-1)/(3\alpha-1)$, whose blow-up rate is $y \gtrsim (t^* - t)^{-\rho}$, $\rho(\alpha) = 3 - 1/\alpha$. The energy inequality supplies $\|u\|^2 \in L^1_{\text{loc}}$ and nothing more, so a blow-up is refuted precisely when that rate is *not* locally integrable:

$$\neg \text{IntegrableOn } (x \mapsto x^{-\rho(\alpha)}) (0, T) \iff \alpha \geq 1/2.$$

That equivalence is kernel-verified, with the integrability half done in genuine measure theory (Mathlib's characterisation of L^1 for real powers), not as a **Prop**-level placeholder. α enters the entire development at *one* lemma.

Scope, stated in the file and the ledger: this is *not* Cheskidov's Theorem 4.4. It contains no bilinear estimate (an input), no local existence, no Galerkin approximation and no limit passage — which is where the real cost of that theorem sits.

Read backwards, the same theorem quantifies the barrier. Refuting blow-up from a bound $\|u\|^\theta \in L^1_{\text{loc}}$ needs $\theta \geq \theta^*(\alpha) = 2/\rho(\alpha)$, and $\theta^*(\alpha) > 2$ exactly below $\alpha = 1/2$: at $\alpha = 2/5$ one needs $\theta = 4$ and has 2; at $\alpha = 7/20$, $\theta = 14$. The obstruction is a single scalar deficit with exactly two doors — raise θ , or leave the Riccati route — and any future proposal must now name its door and its gain before implementation. That is the pre-registered screen the four dead mechanisms of §10 lacked.

12.1 The screen, and a control broken by its own run

Before searching for a better bound, the cheap question is whether one is even plausible. The probe measures $\int_0^T \|u\|^\theta$ as a function of the shell truncation N : a finite truncation cannot blow up, so a single N proves nothing, and the signature of non-integrability is growth without saturation.

Its theorem-backed positive control failed on the first run — reporting divergence at $\alpha = 1$, where regularity is a published theorem. The cause was neither the integrator nor the observable: the seed $u_n(0) = A2^{-\beta n}$ has finite enstrophy only for $\beta > \alpha$, and at $\alpha = 1$, $\beta = 0.7$ the initial datum has *infinite* enstrophy in the limit. The theorem's hypothesis $u_0 \in V$ was violated, and the measured N -dependence belonged to the initial data rather than the dynamics. The same failure exposed a second defect: runs that stopped early accumulated over *different* time windows, making a diverging control look as though it were saturating. Both are now structural — the harness refuses to run when $\beta \leq \alpha$, and all runs are compared over a common window (LL-17).

After repair the controls behave: $\alpha = 1$ saturates to three decimals, and $\alpha = 1/4$ with large positive data diverges with final times collapsing from 2 to 0.018 — unambiguous blow-up. At moderate amplitude both data classes saturate at $\alpha = 2/5$, but that is the small-data regime where regularity is trivial and saturation carries no information.

The discriminating run, and the third control to save a wrong conclusion

The measurement that matters is at large amplitude, and there the comparison is theorem-backed on one side: Barbato–Morandin–Romito proves positive data at $\alpha = 2/5$ globally regular at *every* amplitude, so that column must saturate however large the data.

It did not. At $A = 8$ and $A = 32$ the positive column showed the same apparent blow-up signature as the sign-changing one — final times collapsing from 2 to 0.08, growth ratios far from 1. Since the positive case is a theorem, the signature had to be an artifact, and it was: the runs were terminated by the integrator’s *step-count cap*, not by its magnitude guard. Shrinking final times meant the computation ran out of budget. **The entire amplitude sweep is inadmissible as evidence**, in both columns.

The harness now separates the two stop reasons explicitly and refuses to let a compute-limited block be read at all. This is the third occasion in this campaign where a pre-registered control turned a publishable-looking number into a caught artifact — after the σ null-model catch (§11) and the mis-stated regime band (§14, LL-16). The pattern is consistent enough to be the campaign’s main methodological finding: in this domain, the artifact rate for uncontrolled measurements appears to be close to one.

The open question therefore stands exactly where the **Tier A** barrier leaves it, with no numerical evidence in either direction. Rather than build a better integrator, the next step changed the observable and attacked the barrier’s first door algebraically, where this programme’s instruments are exact rather than approximate.

12.2 Door #1, closed: there is no banded quartic invariant

The barrier needs L^1_{loc} control of $\|u\|^\theta$ with $\theta \geq 4$ at $\alpha = 2/5$. Now $\|u\|^2 = H_\alpha$ is *quadratic* in u and $\|u\|^4 = H_\alpha^2$ is *quartic*, while energy methods produce L^1 control of quadratics — that is what they are. So door #1 reduces to one algebraic question, and it has an exact answer.

Identity (Q1), and what it settles

For $H_\gamma = \sum_n \lambda^{2\gamma n} u_n^2$ on the truncated system,

$$\frac{d}{dt} H_\gamma = -2\nu H_{\gamma+\alpha} + 2(\lambda^{2\gamma} - 1) \sum_n \lambda^{(2\gamma+1)n+1} u_n^2 u_{n+1}.$$

The nonlinear prefactor vanishes **only** at $\gamma = 0$. Hence *energy is the unique conserved weighted quadratic*: $\theta = 2$ is the entire quadratic supply, not an artifact of technique, and door #1 genuinely requires leaving degree 2. (Q1) generalises this repository’s own **Tier A shellB_energy_conservation**, which is its $\gamma = 0$ case. A second reading matters as much: for $\gamma < 0$ and *positive* data the prefactor is negative and the sum non-negative, giving a monotone family — which is exactly where positivity does its work in the literature, and exactly what is unavailable for sign-changing data.

An exact rational search for conserved quartics of the inviscid truncated system — diagonal $\sum c_n u_n^4$, neighbour $\sum c_n u_n^2 u_{n+1}^2$, and all banded quartics of index spread below w , up to 75 monomials — returns **nullspace dimension zero in every case**. The pre-registered criterion is met: **door #1 is closed for banded polynomial quartic conserved quantities**, the natural home of an energy-method improvement. Not even E^2 survives, since it is not banded.

What the closure does *not* reach, and what must travel with any citation of it: quartics that are monotone without a polynomial certificate, non-polynomial quantities, quantities conserved only on invariant subsets, and door #2 — leaving the Riccati route altogether. A closed door is not a closed problem.

The search’s own control history is worth one line, because it was informative rather than merely corrective. Its first negative control *failed*: perturbing the in-flux exponent $\lambda^n \rightarrow \lambda^{n+1}$ does not destroy conservation at all, it merely moves the conserved weights from $c_n = 1$ to $c_n = \lambda^{-n}$. The telescoping is robust to the exponent and sensitive to the *index structure*. That perturbation now serves as a second positive control, and the genuine negative control breaks the index structure instead.

13 The 3-D Bridge Continued: the Nonlinearity, the Energy Identity, and the Helical Basis

This section reports the work of 9–10 September 2026. Four results are new, and three of them are *negative*: they remove claims the programme had been carrying. That ratio is not an accident of a bad week; it is what the gating system is for, and reporting it is the point.

13.1 The Leray-projected nonlinearity, and a theorem that was true but empty (Tier A)

`FourierDynamicsZ3.lean` defines the Galerkin index set $\Lambda_M = \{k \in \mathbb{Z}^3 : |k|^2 \leq M^2\}$ as a `Finset`, the convective term, and the operator

$$B(u, v)_k = \mathbb{P}(k) \left[-i \sum_{p+q=k, p \in \Lambda_M} (q \cdot u_p) v_q \right], \quad (13)$$

which is the *corrected* convolution formula certified in exact arithmetic in §9. Proven: transversality $k \cdot B(u, v)_k = 0$ unconditionally, and the triad-support property that `sublattice_invariance` takes as a hypothesis — so that theorem now applies to a concrete operator rather than a hypothetical one.

That last point had been flagged as audit item F1, and it was not pedantry. The invariance theorem is conditional on a hypothesis *the zero map satisfies trivially*: it was true, and possibly about nothing. The repair is an explicit witness. With $k_0 = (1, 0, 0)$ carrying amplitude $(0, 1, 0)$ and $q_0 = (0, 1, 0)$ carrying $(1, 0, 0)$, one computes $B(\delta_{k_0} a, \delta_{q_0} b)_{k_0+q_0}$, first component, $= -i/2 \neq 0$. Two negative controls fire: perturbing the computed value to $-i/3$ breaks the proof, and asserting the same existence at $M = 0$, where the mode leaves the ball, breaks it too.

Audit item F4 is closed in the same file. The ball is built cube-then-filter while the Galerkin cutoff is stated by squared norm; `mem_ball_iff` shows they agree, the non-obvious direction being that $|k|^2 \leq M^2$ *already* forces every coordinate into $[-M, M]$, since each k_i^2 is one non-negative term of the sum. The cube is implied, not imposed.

13.2 The energy identity is a two-swap, and its abstract form was already proved here

The identity $\text{Re} \sum_k \langle u_k, B(u, u)_k \rangle = 0$ had been scoped as hard and left as a `Prop` for two weeks. It is one symmetry argument.

Write the triad sum with $r := -k$, so the constraint reads $p + q + r = 0$. The factor $(u_q \cdot u_r)$ is *symmetric* in q, r while the constraint set is invariant under exchanging them, so averaging the two orderings gives

$$2S = \sum_{p+q+r=0} ((q+r) \cdot u_p) (u_q \cdot u_r) = - \sum_{p+q+r=0} (p \cdot u_p) (u_q \cdot u_r) = 0, \quad (14)$$

by divergence-freeness at p — and by nothing else. No reality condition, no Leray projection, no truncation convention enters.

Two things are worth recording about *why* this took two weeks. First, the exact-arithmetic harness had noted that “a first attempt at a direct 3-way relabeling argument did not close

cleanly”. The diagnosis is that a *3-cycle* cannot close, because p is distinguished: it is the index whose amplitude sits inside the wavevector contraction. The 2-swap that *fixes* p closes in one line. The failure was of the chosen group, not of the identity. Second, and more embarrassing, `AbstractAlgebraicConservation.triad_sum_zero` — whose `swap3` is exactly that two-swap — had been **Tier A** in this repository since 13 August. Task 2.2 is therefore an *instantiation* obligation, not a proof obligation.

A *reuse audit* is now a required stage of the formalisation workflow (`docs/designs/FORMALIZATION_WORKFLOW`). The same document records why the common habit of stubbing a hard lemma with `sorry` is wrong here: Gate 2 fails on any `sorry`, and a `sorry`’d theorem still *defines its name*, so every downstream `#print axioms` silently carries `sorryAx`. The supported substitute is a `Prop` with an inhabited quantifier domain.

The instantiation, carried out (Tier A). `energy_conservation` now proves (14) for the concrete truncated operator, for every M , with no smallness, no genericity and no hypothesis beyond the Galerkin structure. Four bridge steps carry it: the Leray projector drops out of the outer pairing; the double sum over $\Lambda_M \times \Lambda_M$ is reindexed by $(k, p) \mapsto (p, k - p)$; the terms whose middle member has left the ball are discarded, which the cutoff supplies; and the abstract theorem applies. The third step is the one the design memo flagged as the only real trap, and it is real: the set the reindexing lands on is *not* closed under the two-swap, because it constrains p and r but not q . The cutoff repairs it, and the repaired set is closed for the reason that matters — its defining condition is a property of the *unordered* triad, which the swap merely permutes.

One deviation from the memo simplifies the proof. The memo routes the projector step through the conjugate symmetry $\overline{u_k} = u_{-k}$; that is not needed, since integer wavevector entries are fixed by conjugation, so divergence-freeness transfers to $\overline{u}$ directly and the step holds for *any* divergence-free field. Conjugate symmetry is still used, once, and elsewhere: to turn the Hermitian outer pairing into the bilinear form the abstract theorem is stated for.

The three negative controls the memo mandates all fail as required. The sharpest supplies divergence-freeness at the *wrong* index of the pair and does not typecheck — which is exactly what a 3-cycle would have needed, so its failure is the formal counterpart of why the first attempt stalled.

Audit verdict D1 still binds, and the scope should be stated plainly: this is the *finite-dimensional* identity. The Galerkin ODE conserves energy exactly, which is a prerequisite for global existence *of the truncated system*. It is not evidence about the $\alpha' \rightarrow 0$ limit, about Hypothesis U, or about Navier–Stokes.

13.3 The weighted identity: energy conservation and vortex stretching are one theorem

Energy is not what Hypothesis U is about. The specification states it for the *enstrophy* $\|\nabla u\|_{L^2}^2$, which on the lattice is $\sum_k |k|^2 |u_k|^2$, so the quantity that matters is the production

$$P = \operatorname{Re} \sum_k |k|^2 \langle u_k, B(u, u)_k \rangle,$$

the same sum as (14) carrying a weight. Whether the two-swap kills that too is not an open question — it does not, or three-dimensional Navier–Stokes would be easy. The useful question is *how* it fails, and that has an exact answer.

The pair identity (Tier A). Write $r = -(p + q)$, and let w be *any* weight, attached to the third member since $k = -r$. The two orderings of a triad differ in the weight and in nothing else: $\operatorname{dot}(u_q, u_r)$ is symmetric, and divergence-freeness at p gives $\operatorname{dot}(k_r, u_p) = -\operatorname{dot}(k_q, u_p)$. Hence

$$w(r) (k_q \cdot u_p) (u_q \cdot u_r) + w(q) (k_r \cdot u_p) (u_r \cdot u_q) = (w(r) - w(q)) (k_q \cdot u_p) (u_q \cdot u_r). \quad (15)$$

Divergence-freeness is used exactly once, and the whole reduction turns on it; a control that removes it breaks the proof.

What (15) says. Constant w makes the right side vanish, so *energy conservation is the constant-weight case* — and that is machine-checked rather than asserted, since the energy corollary is derived *from* (15) using nothing about the value of the constant. The obstruction is carried entirely by the weight difference across the swap: not by the triad’s geometry, not by chirality, not by the lattice, because every other factor is common to the two orderings. With $w = |\cdot|^2$, instantiated for the concrete truncated operator,

$$2 \sum_k |k|^2 \langle u_k, B(u, u)_k \rangle = -i \sum_{\text{triads}} (|r|^2 - |q|^2) (q \cdot u_p) (u_q \cdot u_r). \quad (16)$$

This is vortex stretching in closed algebraic form: a triad transfers enstrophy in proportion to how *unequal in wavenumber* its two swapped members are, and a triad whose two swapped members share a sphere transfers none. Set (16) beside (14), which is the same computation with the weight removed and whose right side is therefore zero, and the entire difference between the two conservation questions is one factor. The negative control that matters replaces that factor by zero — asserting enstrophy conserves as energy does — and fails.

What it is not, which is the important half. It is an *identity, not a bound*: it gives no estimate on P . It says nothing about uniformity in the truncation, which is the entire content of Hypothesis U, and obstruction O5 stands untouched — at fixed truncation the system is regular by an elementary argument, so a result that does not use the limit uniformly proves nothing about the limit, and this one does not use the limit at all. Nor is it new mathematics; the enstrophy production of the Fourier–Galerkin system is classical. What is new is only that it is kernel-checked in the same abstract form as the energy identity, so the two are visibly one theorem. It unblocks nothing: the roadmap’s remaining items stay blocked on definitions that have not been authored.

A procedural note belongs with it. Unlike the energy identity, whose design memo was written by the programme owner and executed here, the memo behind (15) is self-authored. Its proofs are machine-checked and its arithmetic is exact, but whether the *statement* is the one the programme wants is a human judgement, and it is precisely the judgement neither gate can make. It is carried as a draft pending that audit.

13.4 The regularized system is the truncated system, and its balance laws close

A framing fact deserves its own subsection, because it connects everything in this section to the programme’s title. The specification identifies the T-dual regularization at the PDE level with the frequency projection onto $|k| \leq 1/\sqrt{\alpha'}$. *So the Galerkin system formalised above is already the α' -regularized system*, under the correspondence $M \leftrightarrow 1/\sqrt{\alpha'}$, and Hypothesis U is exactly the statement that the enstrophy bound is uniform in that correspondence — nothing else. No separate “regularized operator” remains to be defined, which matters under the programme’s rule that definitions may not be invented downstream.

With that identification, the fixed-truncation bookkeeping is now complete in the kernel (**Tier A**). Writing $F_k = -\nu |k|^2 u_k + B(u, u)_k$ for the right-hand side of the regularized system, in the same algebraic-rate sense as the dyadic laboratory — the rate is the pairing of the state against the right-hand side, with no time variable and no ODE:

$$\operatorname{Re} \sum_k \langle u_k, F_k \rangle = -\nu \sum_k |k|^2 \|u_k\|^2 \leq 0 \quad (\nu \geq 0), \quad (17)$$

$$\operatorname{Re} \sum_k |k|^2 \langle u_k, F_k \rangle = -\nu \sum_k |k|^4 \|u_k\|^2 + \operatorname{Re} \sum_k |k|^2 \langle u_k, B(u, u)_k \rangle. \quad (18)$$

In (17) the nonlinearity contributes nothing, by the energy identity, so the energy of the regularized system can only fall. In (18) the production term is the vortex-stretching sum of (16), exhibited nonzero on a genuine Galerkin state. **Nothing bounds that term against the dissipation uniformly in M ; that absence is precisely Hypothesis U.** The negative control that matters here drops the production term from (18) — asserting that enstrophy dissipates as energy does — and fails, so the balance law displays the open problem rather than hiding it.

This is where the formal programme now stands: everything at fixed truncation that the roadmap asked for is kernel-checked, and every further step is either classical PDE consolidation on paper or the uniformity mechanism itself, which is mathematics that does not yet exist.

13.5 A “frustration index” that measures its own null model (Tier C, refuted as posed)

A programme narrative proposed a triadic frustration index $\mathcal{D}(M) = \sum |T| / |\sum T|$ over the Galerkin ball, predicted it to grow like M^3 , and read that growth as the lattice stifling the ultraviolet cascade. It was computed to $M = 20$ in Rust, against three families of fields on the same lattice with the same spectral envelope.

field	slope, flat envelope	slope, Kolmogorov envelope
random independent phases (<i>the null model</i>)	3.43	1.89
phase-coherent	−0.12	−0.20
deterministic arithmetic construction	4.01	2.07

The growth belongs to the *random phases*, not to $\mathbb{Z}^3$: a phase-coherent field gives $\mathcal{D}$ flat in M . The predicted M^3 is the null model’s own arithmetic — a random walk over the same number of terms. Two of the three coherent comparators are moreover degenerate: one has zero net flux across every sphere, and one has identically zero transfer by parity, discovered only because a liveness guard was added after an inert control passed silently. $\mathcal{D}$ is not a robust instrument, and directive D-2 accordingly replaced global scalar ratios with *local directional fluxes* — net helicity transfer into named shell bands.

13.6 A companion solver’s “Beltrami shield”, and a control that refuted our own review

A companion dual-scale solver reported that a T-dual metric converts a finite-time blow-up into a stationary Beltrami flow. A source-level audit found the reported alignment to be a monotone function of a hand-added damping term on the negative-helicity amplitudes, the Lamb-vector norm to be *derived from* that same alignment rather than computed from $u \times \omega$, the frustration index there to return ∞ on a frozen flow because only its denominator is tested, and the blow-up threshold to be crossed by an ordinary cascade into a 20-shell truncation, where $\Omega \leq \kappa_N^2 E \approx 2.7 \times 10^{11} E$ against a threshold of 10^6 .

The decisive control was then run, and *it partly refuted that audit*. The instruction as posed — “damping on, $\alpha' = 0$ ” — cannot be executed, because in the source the damping is *derived from α'* ; zeroing it zeroes the damping too. A 2×2 factorial separating metric from damping gives:

run	metric	damping	alignment	outcome
A	k_{eff}	on	0.9888	bounded
B	bare k	on	—	DIVERGED, inadmissible
C	k_{eff}	off	0.9805	bounded
D	bare k	off	—	DIVERGED, inadmissible

Boundedness is supplied by the *metric*, not by the damping: the metric alone stays bounded while the damping alone diverges. But the sharper finding is that the initial condition is *already* 0.9560 aligned by construction, so the claimed “topological conversion into a Beltrami flow” is a move of +0.0329 from a state chosen to be 95.6 % Beltrami. The owner recorded the spontaneous Beltrami attractor as KILLED.

Two defects in our own control were caught by the control itself: its first run interpreted a *diverged* cell, and its negative perturbation was *inert*, indexing the same shell on both branches. Both are fixed and recorded.

One further correction belongs here because it was nearly built upon. The directional-flux instrument reported *exactly zero* net transfer into the ultraviolet band under the metric, and an analytic explanation was requested. There is none to give: the band is never populated. Peak amplitudes fall 2.6×10^{-9} at shell 8, 1.5×10^{-43} at shell 10, 3.6×10^{-184} at shell 12, and below the smallest representable double thereafter. The zero is an *underflow*. The real effect — super-exponential suppression of the cascade — is genuine; an algebraic cancellation to explain a number that is really $\sim 10^{-370}$ would be a ghost.

13.7 The helical basis, the exact triad coefficient, and a condition with no content

With the two analytic tracks blocked on definitions that may not be invented, the remaining tractable branch is Waleffe’s helical decomposition, whose content is *algebra*: the interaction coefficient is a determinate function of three wavevectors and three signs.

The coefficient, derived and then corrected by test. Projecting the Leray-filtered non-linearity onto $h^\pm$ and antisymmetrising — the same two-element move as (14) — gives $C = -\frac{1}{4}(s_p|p| - s_q|q|)g$ with $g = (h^{s_p}(p) \times h^{s_q}(q)) \cdot \overline{h^{s_k}(k)}$. In the triad’s own planar frame this collapses onto a single geometric scalar:

$$C_{k,p,q}^{s_k s_p s_q} = +\frac{i S_{pq}}{4|k|} (s_p|p| + s_q|q| + s_k|k|)(s_p|p| - s_q|q|), \quad S_{pq} = (\hat{\kappa}_p \times \hat{\kappa}_q) \cdot \hat{n}. \quad (19)$$

The first derivation of (19) was *wrong*: it carried $-s_k|k|$, because the reduction of the signed areas used the relation for a triad written $p + q + r = 0$ while the convention in force was $p + q = k$. A triad-by-triad test against a brute-force construction exposed it immediately — all eight magnitudes matched but were *permuted*, the signature of a sign inside a factor. The corrected form agrees with brute force to 2.1×10^{-16} in all eight chirality classes, and it authenticates itself: the balance factor becomes the sum of *all three* signed helical wavenumbers, i.e. the classical Waleffe resonance condition, which the erroneous version did not reproduce.

The resonance condition is exactly collinearity (Tier A). Equation (19) has three vanishing loci. The middle one is empty of content:

Theorem 13.1 (Resonance = collinearity). *For a triad $p + q = k$ of nonzero real vectors, $s_k|k| + s_p|p| + s_q|q| = 0$ holds for some choice of signs if and only if the triad is collinear.*

Proof. The magnitudes are positive, so the signs cannot all agree; exactly one differs and the condition reads $|k| = |p| + |q|$, or $|p| = |k| + |q|$, or $|q| = |k| + |p|$. Each is equality in the triangle inequality for $k = p + q$, which holds exactly when the two vectors on the right are parallel and like-directed. The converse is immediate. $\square$

Collinear triads already carry no transfer, for want of a plane. *So the resonance condition selects no new inert triads.* The theorem is kernel-verified in `HelicalBasis.lean` via Cauchy–Schwarz through Lagrange’s identity, every step after one squaring being integer arithmetic; it

was first established in exact integers over 558 090 lattice triads by two independent decision procedures agreeing without a single disagreement.

Note what the proof does *not* use: $\mathbb{Z}^3$. The programme’s recurring appeal to “the arithmetic rigidity of the lattice” contributes nothing here — this is the triangle inequality, true for real vectors. That is the third independent deflation of the same narrative, after §13.5 and the ultraviolet underflow of §13.6.

What survives, and the basis that carries it (Tier A). The non-degenerate content is the *other* factor: $C = 0$ whenever $s_p|p| = s_q|q|$. This is immediate from the antisymmetrisation, and unlike the resonance condition it *bites*: it is satisfied by every pair of lattice modes sharing a sphere with equal chirality, and a witness records a non-collinear instance, so it is not a restatement of the degenerate case.

`HelicalBasis.lean` proves the five basis facts — $h \cdot h = 0$, the normalisation, transversality, the curl eigenvector property $i k \times h^s = s|k|h^s$, and $\bar{h}^s = h^{-s}$ — with the choice of $\nu(k)$ made *total* and its degeneracy excluded on every nonzero lattice point. Two design decisions are worth recording. The vector is kept *unnormalised*: $|k| = \sqrt{k_{sq}}$ is irrational at almost every lattice point, so a normalised basis drags a square root and a division through every proof, and in Lean an unwitnessed division silently proves a different theorem, since $x/0 = 0$. And the basis is *parametrised* by ν , because (19) is derived in the triad’s own planar frame and a basis hard-coding one ν cannot state it — a generic ν changes C by a phase, as the crucible measured. On the lattice that planar frame is available as an *integer* vector, $p \times q$, orthogonal to p , to q , and — since $k = p + q$ — to k as well; and it degenerates exactly on the collinear triads, that is, by Theorem 13.1, exactly where the coefficient has no content anyway.

13.8 The closed form in the kernel, and the converse that closes the inventory

A gap between tiers, found and closed. Theorem 13.1 is stated “for some choice of signs”, and the exact-arithmetic sweep that first established it quantified over all three relabellings. The first Lean transcription did not: it assumed $-|k| + |p| + |q| = 0$, the single pattern in which k is the odd one out. The two tiers were therefore asserting different propositions, and the machine-checked one was the weaker — a failure mode worth naming, because it is invisible to both gates individually. The remaining two patterns are now proved by rewriting $p + q = k$ as $k + (-q) = p$ and $k + (-p) = q$, and the general statement quantifies over $s \in \{\pm 1\}^3$: of its eight branches, two are impossible because a sum of three positive reals is not zero, and the other six are the three patterns, each arising twice under global sign flip.

The closed form, verified in three independent ways (Tier A, Tier B). Equation (19) is now kernel-compiled. In the triad’s own integer frame $N = p \times q$ the memo’s S_{pq} and $|N|^2$ coincide, so in the unnormalised scaling that keeps every proof free of square roots and divisions it reads

$$g = -i (N \cdot N)^2 (s_p|p| + s_q|q| + s_k|k|), \quad C = \frac{i}{4} (N \cdot N)^2 (s_p|p| + s_q|q| + s_k|k|) (s_p|p| - s_q|q|). \quad (20)$$

The proof needs no $i^2 = -1$: every term quadratic in i is a triple product $(N \times x) \cdot N$, identically zero, so i survives only linearly and the whole identity is a degree-9 polynomial identity in six integer variables.

The kernel certifies the *proof*, not the *statement*, and a mis-transcribed sign would be proved just as happily — which is exactly what happened to the first draft of (19). So (20) is also transcribed independently and checked in *exact integers*. This is possible, despite $|p|, |q|, |k|$ being irrational at almost every lattice point, because the closed form is linear in them and the brute force at worst quadratic: both sides are polynomials in three formal symbols with integer coefficients, and polynomial equality is integer equality. The identity is thereby confirmed *in the*

magnitudes, over 5 256 triad-by-chirality-class checks, rather than at the particular magnitudes those triads happen to have. Three negative controls fail as required, the sharpest being an orientation flip to $q \times p$: $N \cdot N$ is blind to it and g is not.

The converse, and why it is the result that matters. Every vanishing statement to this point has been one-directional. None excluded a fourth, unnoticed locus, so the programme had no way to know when its inventory of inert triads was complete. Since $\mathbb{C}$ has no zero divisors, (20) settles it: $C = 0$ if and *only* if the triad is collinear, or resonant, or balanced. With Theorem 13.1 folding the second into the first, *the inventory is closed and has two members* — a degeneracy that carries no transfer for want of a plane, and the balance condition $s_p|p| = s_q|q|$, which is the only non-degenerate one and is witnessed non-collinearly.

Two further remarks. The vanishing theorems of the preceding subsection were proved *geometrically*, by degenerating the frame; (20) re-derives all three *algebraically*, from an identity that never mentions a degenerate frame, so the agreement of the two routes is a cross-check the kernel now runs on every build — and precisely the check the erroneous first draft would have failed. And the deflation of §13.7 is sharpened rather than softened: the closed form gives the exact size of the transfer everywhere off the two loci, and nothing in it distinguishes $\mathbb{Z}^3$ from $\mathbb{R}^3$.

13.9 The gauge separation, and a mechanism retracted

The helical vector depends on an arbitrary transverse choice $\nu(k)$, so a claim about the *phase* of C is a claim about a gauge. One item was carried as open: the planar frame of (19) is chosen per triad and is therefore not a global gauge, and the step “the signed sum may be evaluated triad by triad in each triad’s own frame” was recorded as observed but not proved. It was described as small and off the critical path. It was neither: closing it retracted the mechanism that had been offered for a measured cancellation.

The two frames respond oppositely to negation (Tier A). The mechanism read: *in a triad’s own planar frame the coefficient is purely imaginary, so $\overline{C} = -C$ and each triad is cancelled exactly by its negation*. Both halves are true, and they hold in different gauges. A global ν satisfies $\nu(-a) = -\nu(a)$, so its frame *flips* and negation *conjugates* the coefficient. The triad frame $p \times q$ is quadratic, so $(-p) \times (-q) = p \times q$: it is *fixed*, and negation leaves the coefficient *unchanged*. The separation is total rather than marginal — in an exact sweep of 48 triad-by-class cases the conjugation law fails in 48 of 48 in the triad frame and 0 of 48 under a global ν , and the invariance law does the exact reverse. So the asserted cancellation occurs in neither gauge: in the triad frame a triad and its negation sum to *twice* a coefficient that a lattice witness shows is nonzero, and under a global ν the pairing projects the sum onto its real part, nonzero in 40 of 48 cases.

A control refuted the natural explanation, and the theorem is stronger for it. Pure-imaginaryity looks like it should come from $p \times q$ being orthogonal to the triad plane. A control moved the common normal off the plane, expecting the real part to appear; it stayed exactly zero in all 48 cases. The actual cause carries no geometry: every term that could hold an even power of i is a triple product $(N \times x) \cdot N$, which vanishes for any N whatever. The kernel statement therefore has *no hypotheses* — the coefficient is purely imaginary whenever the three helical vectors share one frame vector — and it explains why the argument cannot be repaired by choosing a better frame. What breaks it is using *three different* frame vectors, which is exactly what a global ν does.

What is retracted, and what stands. The measurement is not contested: the per-class signed sum is numerically zero under two independent ν constructions. Only the explanation is withdrawn. Under a global ν the negation pairing does not cancel triads, it projects the sum onto

its real part, so the observed zero requires the real parts to sum to zero over the ball — a fact the retracted argument does not supply, and naming the true mechanism is open. The earlier verdict is untouched and mildly reinforced: a zero produced by a symmetry of a negation-symmetric lattice carries no information about turbulence, and it now looks less specific to the physics than when it was first read.

13.10 The symmetry kernel of the production sum, mapped the hard way

With the production sum in exact closed form, the adjudicated attack asks whether it cancels for generic states faster than a random-phase null — *phase mixing*. A pre-registered measurement was built: the ℓ^1 cancellation ratio of the sum over the ball, exact rational end to end (the sum is polynomial in the amplitudes, so no float is ever needed), with deterministic state families, twenty seeded nulls, and a coherence control required to pass before anything is interpreted. The control tripped three times, and each trip became a theorem.

Trip one. The all-real coherent family came out exactly zero. Mechanism, kernel-checked the same day: real amplitudes plus conjugate symmetry make the state parity-even, and the production terms cancel pairwise under global negation (**Tier A**). In passing, the same pairing shows the real part of the term sum vanishes for *every* conjugate-symmetric state — which is precisely why the physical production is real — and that invariant is now asserted by the instrument on every run.

Trip two. A non-real aligned phase came out exactly zero as well, and parity does not explain it. The construction gave every mode's amplitude one fixed director d ; for such states each triad's six orderings collapse to $2c^3 \det[a, b, d] \cdot E$ with $E = 2(ad)(bd)(B-A) + (bd)^2(B-C) + (ad)^2(C-A)$, and the observed zero reduces to the identity

$$\sum_{\text{triads of ball } M} \det[a, b, d] E = 0 \quad \text{identically in } d,$$

verified exactly at $M = 2, 3$ (all ten coefficients of the cubic form vanish, over 91 and 1178 triads) and proved for every M by equivariance: the ball is invariant under signed permutations, the summand transforms with the determinant sign, so each axis reflection forces every monomial odd in each variable and a transposition kills the lone survivor $d_0 d_1 d_2$. Removing a single point from the ball breaks the identity — the cubic symmetry is load-bearing. Notably, the stratum does *not* vanish per triad, and random-phase director states have genuinely nonzero production: only *alignment* collapses onto the doomed sum.

Trip three. With directors varied per mode — the registered repair — the tilted coherent control still fell to the null's level and below it at $M = 3$. This time the mechanism is structural in the observable itself: the sum is trilinear, every triad splits its members two-and-one across the conjugate halves, and negation sends each term to minus its own conjugate; together these force *any* single-phase family to collapse to $T = 2i \operatorname{Im}(cX)$ for a single c -independent number X . Verified exactly: X solved from two phase tilts predicts a third tilt's sum to the last digit, and X is purely real at both truncations. So the registered control-family *class* was structurally incapable of certifying coherence — a corollary being that the ratio is blind to any global phase, so coherence for a trilinear observable can only mean *relative* phase structure across modes. Per the registration, the observable itself is now at the programme's review gate, with redesign candidates (chirp phases, two-phase interference sweeps, windowed ratios, evolved-state ensembles) awaiting decision; the null comparisons remain quarantined.

Three trips, three mechanisms — parity, cubic equivariance, trilinear collapse — and one standing lesson, now three for three in this programme: an unexplained exact zero is a symmetry not yet found, and it must be found before anything else is believed.

13.11 The external formal landscape, audited (Tier L, proposed)

A publicly released Lean 4 development attributed to OpenAI states and proves the *negative* directions of the Clay problem’s menu: forced Navier–Stokes breakdown on $\mathbb{R}^3$ and on the torus (options C and D, with a force constructed as the residual of a pre-built singular candidate), and unforced Euler blowup for a constructed compact smooth datum with a quantitative singularity package. This programme audited the tree under its own rules before citing a line of it: a full rebuild (517 modules, zero errors, Lean 4.34.0-rc2), then axiom footprints on the four headline theorems — all exactly `{propext, Classical.choice, Quot.sound}` — with a built-in negative control (the repository’s four intentional challenge placeholders, adapted from DeepMind’s formal-conjectures statements, must and do show `sorryAx`), and an import scan proving the proof tree never touches the placeholder modules. One structural caveat travels with any citation: placeholder and proof share fully-qualified names in modules that can never be co-imported, so the equivalence of the proved statements to the challenge statements is a human side-by-side audit, not a kernel fact. The asymmetry matters more than the results: everything verified there concerns constructed data or manufactured forces; nothing touches Statement A, global regularity, or the uniformity question this programme has isolated — the negative directions are now machine-checked territory, and the open problem is exactly where it was.

13.12 The kinematic track killed with positive evidence, and the pivot’s first exact contact

The review gate of §13.10 was decided by an external adjudication: kinematic phase mixing was declared a dead end on the argument that adversarial phase-aligned states will always maximize the disparity factor, and bounds must come from dynamics restricting access to such states. That argument is itself a testable claim, and it was tested rather than assumed. Purely imaginary amplitudes make a state parity-odd — the exact complement of the proved parity stratum — and every production term then collapses to a sign product times an integer, so the cancellation ratio becomes a pure integer optimisation over sign vectors, searchable exactly. Greedy alignment reaches ratios of 0.15, 0.19, 0.24 at $M = 2, 3, 4$ against nulls of 0.036, 0.017, 0.008: the null’s cancellation improves as the ball grows while the adversarial ratio rises, the gap widening from $4\times$ to $28\times$. Every such state is a genuine Galerkin state, random signs in the same construction stay at the null, and the coherence control that had tripped three times passes at last — on states that kill the track. No kinematic bound uniform in M can hold. The registered redesign candidate, chirp phases, failed at every M and is recorded as such.

The pivot’s first object is now formal (**Tier A**): the *disparity decomposition*. Grouping the production sum by the integer $\delta = |r|^2 - |q|^2$ partitions it into fibers P_δ , and the $\delta = 0$ fiber — same-shell advection, the discrete face of Kraichnan sweeping — vanishes identically, whatever the amplitudes. Sweeping cannot produce enstrophy by the algebra alone; the dangerous production is non-local in scale, at large $|\delta|$, and “how much comes from large disparity” is now a well-posed question.

The first *dynamical* contact is exact and short: forward Euler on Gaussian-rational states, with a scheme-fidelity criterion added after the first attempt showed the integrator injecting energy at 11–100 times the physical dissipation (those runs are disqualified by their own criterion, which is the control on it). In the qualified weakly-nonlinear regime the energy-step identity — a dynamic regression of the energy theorem — held exactly at every step, both step sizes agree to four decimals, and both registered hypotheses are *false*: from the adversarial state the ratio rises monotonically, and from a random-phase state it rises by a factor of forty. The dynamics does not scramble triad alignment on this horizon; it generates it, as nonzero energy transfer requires. One candidate definition is thereby retired before any proof was attempted against it: an invariant region cannot be a neighbourhood of random-phase states, and the right observable along trajectories is the production-to-dissipation ratio, not the cancellation geometry. Longer horizons are not reachable exactly and belong to floating-point exploration calibrated on this run.

None of this bears on the limit or on Hypothesis U; obstruction O5 stands.

13.13 The Core–Tail turn: a calibrated scout, and the transient it measures (Tier C)

The adjudication that followed adopted the disparity decomposition and the trajectory ratio $P/(\nu D)$ as the programme’s observables and redirected the effort toward a *computer-assisted* proof: numerically certified bounds on a core $|k| \leq M_{\text{core}}$ in interval arithmetic, and an analytic bound on the tail $|k| > M_{\text{core}}$. One correction was filed with the acknowledgement and is recorded here because it changes the target statement. A tail that “decays super-exponentially assuming the core is bounded” is not a theorem in three dimensions: the tail is *forced* by the core through exactly the disparate triads isolated above, and a bounded core does nothing to that forcing. What is provable is the self-consistent envelope of rigorous numerics — assume $|u_k| \leq Ce^{-a|k|}$ beyond the core, show that $\nu|k|^2$ dominates the forcing by the boxed core and the enveloped tail, and close — with an exponential (Gevrey) rate, constants supplied by the certificate, and one virtue that is precisely the one wanted against O5: the tail inequality never mentions the outer truncation M , so a closed envelope is uniform in M for the solutions it covers. The objects (core box, envelope, per-mode balance, convolution forcing bound, self-consistency inequality) are proposed exactly and await adoption before any Lean is written against them.

The floating-point scout that the certificate will be built from is Rust: pseudo-spectral, `rustfft` with the 2/3 rule at $N \geq 3M + 1$ so that every aliased product lands outside the retained band, `rayon` over the transforms, fourth-order Runge–Kutta, on a CPU-only workstation by directive. Its right-hand side is calibrated rather than asserted: fed the exact rational trajectory of the preceding subsection, both its engines — the FFT and a direct $O(M^6)$ convolution — reproduce all eight steps of both initial conditions to 2.4×10^{-16} relative, and agree with each other to round-off at $M = 4$. Everything the scout says beyond that is read only where a step-halving partner agrees within 2%.

The first registered protocol asks the one scout-level question that bears on uniformity: at fixed initial energy and viscosity, does the *transient* enstrophy amplification of adversarial states grow with M ? Its outcome at $M = 2, 4, 8$ — every long run readable over the whole horizon by step-halving (worst disagreement 1.3×10^{-5}), the transient itself re-read from every-step runs after the registered sampling interval proved coarser than the $M = 8$ dephasing time, an amendment recorded before the numbers were read — is reported as data. The production-positive adversarial state starts above the enstrophy balance — $P/(\nu D_2)$ equals 1.6, 2.8, 4.1 at $M = 2, 4, 8$, with $D_2 = \sum |k|^4 |u_k|^2$ the dissipation the enstrophy balance is decided against — and loses it within a fraction of a time unit: the production changes sign at $t = 0.059, 0.017, 0.004$ respectively, then oscillates with decaying amplitude. The alignment that the greedy search builds is a kinematic coincidence of triad phases, and the nonlinear phase rotation destroys it on a time scale that falls roughly as M^{-2} — set by the fastest triads, whose rate grows as $M^2 |u_k|$ — while the initial excess grows only as about $M^{0.7}$. The transient gain is therefore small — $Z_{\text{max}}/Z_0 = 1.0023, 1.0122, 1.0265$ — and it *grows with* M , but the growth decelerates: $\times 5.3$ from $M = 2$ to 4, $\times 2.2$ from 4 to 8. Three points are consistent with saturation and with a slow power law alike; the fourth point that would separate them needs the greedy alignment on 17 000 modes and is not part of the registered protocol. Two further readings: the production-negative twin of each adversarial state (the same amplitudes with $P \rightarrow -P$ at $t = 0$, related to it by time reversal of the inviscid part) keeps its sign three to ten times longer and *removes* 10% and 24% of Z_0 at $M = 2, 4$ before dephasing, so viscosity breaks the reversal symmetry in the direction that spends coherence to lower enstrophy — the sweeping-suppression question in dynamical form; and once the alignment is spent, all initial conditions at a given M decay alike, the memory of the phase configuration gone from the enstrophy by $t = 1$. What this supplies to the certificate is the shape it must cover: a transient of duration $\lesssim 0.3$ in which the enstrophy exceeds Z_0 by a factor that is small on the points measured and increasing, decreasingly, in M . It supplies nothing about the

limit; O5 stands.

The certificate’s arithmetic was then built and, more usefully, *measured*. It rests on a single primitive — a rigorous upper bound on the nonlinear forcing of one mode — and on a reconciliation of two constraints that look incompatible: rigorous numerics normally means interval arithmetic with directed rounding, which is floating point, while this programme bans floating point from every Tier B artifact. They are reconciled by rounding in exact rationals: $\text{sqrt_upper}(a/b) = (\lfloor \sqrt{ab} 4^n \rfloor + 1)/(b 2^n)$ has square exceeding a/b because $(\lfloor \sqrt{N} \rfloor + 1)^2 > N$ for every integer N , so every irrational the bound needs is replaced by a self-certifying rational and every comparison in the certificate is between two exact fractions. The precision n buys tightness, never validity.

Two things came out of measuring it. First, a structural improvement: on a divergence-free state $p \cdot u_p = 0$, so $q \cdot u_p = k \cdot u_p$ exactly when $q = k - p$ — verified as an exact Gaussian-rational identity at every pair — and the forcing bound may therefore carry the *fixed* factor $|k|$ in place of the varying $|q|$. That is not cosmetic: a weight constant in the summation variable factors out of the self-consistency inequality, reducing it to an unweighted convolution of the envelope against itself, which is the only form in which the envelope’s summability can be used. Second, and less welcome, the bound is loose. Tightness $\|B_k\|^2/\text{bound}^2$ on random states falls as M^{-3} — an artifact of incoherence, since a random sum of $N \sim M^3$ terms has magnitude $\sqrt{N}$ against a bound of N — so the bound can only be measured on coherent states, supported on two wavevectors and their conjugates, where the sum has a handful of terms that can align. There the worst tightness is exactly $1/4$ at $M = 2$ and $1/9$ at $M = 3$, improved to 0.21 at $M = 3$ by the $|k|$ form. Even at its tightest the bound loses a factor of two to three in norm, a loss a self-consistency argument compounds; the envelope parameters must carry that margin, and the certificate will cover correspondingly fewer states. Recording this before any Lean was written against the bound is the entire point of building the arithmetic first.

One methodological note belongs with it, in the spirit of §14. Three of the five negative controls — drop the $|q|$ factor, use $|p|$ instead, round the roots the wrong way — do *not* break the inequality. That is not a broken control but a measurement: each removes a factor smaller than the bound’s own slack, so a true inequality stays true. The control that certifies the checker can fail is the one that quarters the bound, violated on 864 modes at $M = 2$ and 240 at $M = 3$; its constant was fixed after the tightness was measured, because a sensitivity control must be calibrated to the measured slack to be capable of failing at all. The controls that failed to fail are kept and reported, not replaced.

13.14 Three points read twice: ratios, differences, and a pre-registered prediction (Tier C)

The sentence in §13.13 that the transient growth “decelerates” will not survive being checked, and the way it fails is instructive enough to record in full. It was read off the *ratios* of the excess $e(M) = Z_{\max}/Z_0 - 1$, which do fall: $\times 5.3$ then $\times 2.2$. In absolute terms the same three numbers $e = 0.0023, 0.0122, 0.0265$ have increments

$$+0.0099 \quad \text{then} \quad +0.0143,$$

that is, increments that are *growing*; and so are the injection’s ($\Delta Z_{\text{inj}}/Z_0 = 0.0157, 0.0292, 0.0464$, increments $+0.0135, +0.0172$). Saturation requires the increments to turn over, and nothing here has turned over. Fitting the generic approach-to-a-ceiling shape $e(M) = e_\infty - AM^{-\beta}$ — three parameters to three points, so an exact fit exists — returns $\beta = -0.53$: the exponent has the wrong sign, and the only member of that family through these points *diverges*. The honest statement is thus weaker than the one first written: the falling ratios are compatible with saturation, but no saturating form actually fits, and the “consistent with saturation” reading rested entirely on which of two summaries of the same three numbers was looked at. **Everything in this subsection is stated on three points and is superseded by the fourth — see §13.15,**

where the increments turn over and the same fit becomes admissible. It is retained as written because the sequence of readings is the evidence for the method.

There is, moreover, a mundane reason the ratios fall that says nothing about enstrophy. The fraction of the injected enstrophy surviving to the peak, $e/\Delta Z_{\text{inj}}$, is 0.147, 0.418, 0.571, climbing toward a hard ceiling of 1: at small M dissipation consumes most of the injection before the peak, while at larger M the injection is over sooner ($t_{\text{peak}} = 0.022, 0.011, 0.003$) and more of it survives. So e was forced to grow faster than ΔZ_{inj} early and must decelerate to the injection's own rate as that fraction saturates — which is largely done. The deceleration is, to a good approximation, this bookkeeping.

Accordingly the fourth point was *predicted before it could be computed*, while the $M = 16$ alignment was still running, and the prediction committed to version control. Two independent routes agree: the divergent form $e \approx c(\log_2 M)^2$ with c fixed at $M = 8$ gives $e(16) = 0.047$ and reproduces the growing increments; the survival-fraction bracket gives 0.040–0.052. Hence

$$Z_{\text{max}}/Z_0|_{M=16} \approx 1.040\text{--}1.052,$$

a third consecutive fall in the ratio. **And that is the point of stating it in advance: the registered protocol asks the fourth point to separate saturation from slow divergence, and it is predicted not to.** The divergent model also predicts a third falling ratio ($\times 1.78$) and lands inside the same bracket, so at $M = 16$ the two hypotheses differ by less than the spread of either estimate. They separate at $M = 32$ and cleanly at $M = 64$. What the run retains is value as a falsification test of the bookkeeping model above — a return *outside* the bracket is the informative outcome — and that is a smaller claim than the protocol was written with.

13.15 The fourth point, and the prediction it falsified (Tier C)

The $M = 16$ alignment converged at sweep 42 and the transient ran the same evening. **The pre-registered bracket was 1.040–1.052; the measured value is 1.0324. The prediction is falsified, below the bracket.** The halving pair agrees over the whole window to 9.95×10^{-5} in E and 6.09×10^{-4} in Z , and — because t_{peak} at $M = 16$ is 0.001, only four samples — the entire series was re-measured on a uniform ultra-fine grid ($\Delta t = 6.25 \times 10^{-5}$, every step). Z_{max}/Z_0 is unchanged at $M = 16$ under $16\times$ finer sampling and $M = 8$ reproduces its archived value to 2×10^{-4} , so the archived sampling was adequate below $M = 16$; only the $M = 16$ *injection* was under-integrated, by 4%.

M	$P/(\nu D_2) _0$	t_φ	Z_{max}/Z_0	t_{peak}	$\Delta Z_{\text{inj}}/Z_0$
2	+1.604	0.059	1.0023	0.022	+0.0154
4	+2.840	0.017	1.0122	0.011	+0.0292
8	+4.115	0.004	1.0267	0.003	+0.0468
16	+5.016	0.001	1.0324	0.001	+0.0503

Every series has now turned over, which is precisely what §13.14 said had not happened and could not be asserted on three points. The increments of $Z_{\text{max}}/Z_0 - 1$ run +0.0099, +0.0145, +0.0057; the injection's +0.0138, +0.0176, +0.0035; the initial excess's +1.236, +1.275, +0.901; and t_φ follows M^{-2} exactly. Redoing the admissibility test of §13.14 on the last three points returns $\beta = +1.347$ for $Z_{\text{max}}/Z_0 - 1$ and +2.330 for the injection — *admissible*, where the same fit on $M = 2, 4, 8$ returned $\beta = -0.53$ — with implied ceilings 1.0361 and 0.0512.

That result is stated here with the loudest caveat in this document, because it is the same shape of evidence that was wrong one point earlier. $\beta = +1.347$ is an exact three-parameter fit to three points, and twelve hours before it the identical procedure on the identical observable returned an inadmissible exponent and a confident divergent reading that the fourth point overturned. It is not a verdict and not a tier promotion. $M = 32$ is the test; O5 stands.

The post-mortem is worth more than the number. The bracket rested on two factors, and exactly one failed: the survival fraction was predicted at 0.63–0.71 and measured 0.6441, while the injection was predicted at 0.0634–0.0738 and measured 0.0503, its local exponent falling not to the assumed 0.50 but to 0.104. **The error was extrapolating a composite when its components were in hand** — the injection is the initial excess times the alignment’s lifetime, t_φ follows M^{-2} exactly, and the excess grows $\times 1.219$ rather than the $\times 1.62$ that this report’s own $M^{0.7}$ description implies. A decelerating driver against a lifetime collapsing as M^{-2} gives a saturating product, and that was derivable from committed data before the run. The standing rule (LL-24) is to pre-register the *factors* of a product observable and multiply them, never the composite.

This is what the pre-registration was for. It converted a comfortable reading into a test; the test failed; and because the prediction had been decomposed, the failure localised to one factor instead of leaving a vague disagreement to be argued about afterwards. A programme that only records its confirmed predictions is not running the method.

13.16 The fourth point’s real obstacle was a quadratic algorithm, not a machine (Tier C)

Reaching $M = 16$ at all first required two defects in this programme’s own tooling to be found, and both are worth recording because each was invisible while the code produced correct answers.

The registered protocol had *declared* a memory risk: the greedy alignment materialises a triad table of order 10^8 entries, “ ≈ 3 GB”, against a workstation with ≈ 14 GB free. Counted rather than estimated — by FFT autocorrelation of the ball indicator — the table is 1.367×10^8 entries, so the order of magnitude was right. The byte count was not. One entry occupied 80 bytes as three 24-byte wavevectors, giving 10.2 GB, and the class list was built by a `flat_map` over the whole table *before* its `dedup`, a further 9.8 GB standing simultaneously: a peak near 20 GB, and the run would have died in the allocator with its own memo calling the failure a surprise. Building the table directly in index form — two passes, the wavevector form never materialised, the vector reserved to an exact count — costs 24 bytes an entry, 3.03 GB measured as resident, and removed the wall.

The second defect was larger and had no declared risk attached, because nothing about the output was wrong. The greedy re-evaluated the objective by summing all 1.35×10^8 table entries for *each* of the 8 538 candidate sign flips, every sweep — a cost of $O(\text{classes} \times \text{triads})$ that nobody had questioned in three campaigns, since the answer it returned was correct and the runtime at $M \leq 8$ was seconds. But flipping one class’s sign does not disturb the other terms. Maintaining the signed total S and updating it by

$$S \mapsto S - 2 \sum_{\substack{\text{triads containing} \\ \text{that class an odd number of times}}} s_a s_b s_c g,$$

where the odd-incidence restriction is exactly right because a class occurring twice multiplies its term by $(-1)^2 = 1$, reduces a sweep to $O(3 \times \text{triads})$: a factor classes/3, which is 2846 at $M = 16$ and 22 876 at $M = 32$. The realised gain is smaller than the operation count suggests — random gathers into a multi-gigabyte table are far less efficient per byte than the sequential scan they replace — and is reported as measured, not as the ratio of operation counts.

Both changes are refactorings and were held to a refactor’s standard rather than argued for: the incremental greedy reproduces the archived $M = 8$ run’s nine sweep values digit for digit and its output trajectory *byte for byte*, and at $M = 16$ it reproduces the pre-change run’s sweep-by-sweep objective exactly, the two runs having been kept side by side for precisely that comparison. The same integer arithmetic, the same visit order, the same accept test, therefore the same alignment.

The consequence for the programme is a retraction of a resource estimate this report would otherwise have carried. An earlier costing put the $M = 32$ adversarial alignment at 195 GB and ≈ 108 days of CPU and concluded that no rented machine helps. The conclusion stands

and the reasoning does not: with the incremental objective and an enumeration that reads each class's triads off the lattice instead of storing them, the $M = 32$ alignment needs essentially no memory and an estimated one to two hours. The obstacle was never compute. **It was a quadratic algorithm inside a programme whose stated purpose is rigour, and it survived because correctness gates cannot see cost.** What remains genuinely expensive at $M = 32$ is the trajectory itself, and there a larger machine buys roughly a factor of two — the scout saturates near twelve busy threads and is then memory-bandwidth bound — so the honest ask, if one is ever made, is sixteen cores and not sixty-four.

13.17 The fifth point ($M = 32$): the algorithm fix held, and the prediction it enabled still failed (Tier C)

The fix of §13.16 delivered what it promised: the $M = 32$ alignment converged in 113 sweeps ($9 \rightarrow 42 \rightarrow 113$ across the series — the landscape's depth is growing faster than its size) with essentially no memory pressure, not the 195 GB and 108 days the quadratic costing had implied. This time the prediction was built *factor by factor before the run*, per the standing rule of §13.15: initial excess $[4.85, 5.38]$, dephasing time 0.00025 (M^{-2} exactly), injection $[0.0453, 0.0539]$, survival fraction $[0.663, 0.699]$, giving $Z_{\max}/Z_0 \in [1.0300, 1.0377]$.

Measured: 1.0386, **outside by** $+0.0009$. The miss was localised before the run's own outcome table was consulted: the initial excess came in at 6.604, above its bracket, with a local exponent that *rose* ($0.286 \rightarrow 0.397$) rather than continuing to fall — the third reversal of a monotone-looking trend in two days. Read factor by factor against what actually missed: the injection landed inside its bracket (0.0530), and the survival fraction was the culprit this time (0.728, above $[0.663, 0.699]$), the reverse of $M = 16$, where the injection had been the miss and the survival fraction had been right. **No single factor on this family has been predictable from its own trend across two doublings running.** The admissibility fit that had looked confirmed on four points ($\beta = +1.347$) is inadmissible on five ($\beta = -0.121$); what fits, empirically and *not* as a registration, is a slow power law $e \sim M^{0.27}$.

A second attempt to make $M = 64$ reachable by a cheaper search was tried and is reported as a validated negative result rather than discarded quietly. Dirty-tracking — skip a class's re-check unless something that could change its answer flipped since its last check — has a tight correctness argument and was checked against the *entire* recorded $M = 16$ sweep sequence, all 42 values digit for digit, the converged phases byte for byte. It is **5.5% slower**, because even the converging sweep's handful of flips dirtied over half the class set: the triad co-occurrence structure is close to dense, not locally sparse, and a worklist trick that wins on sparse problems does not apply here. Kept in the tree as a measured negative result (LL-27): a correctness argument bounds what an optimisation is *allowed* to skip, never how much it *will* skip on the object in front of it.

13.18 Breaking the search, twice: an FFT gradient, and a screen that chooses the optimiser instead of the human (Tier C)

$M = 64$ needed a different algorithm, not a faster version of the same one: the exact sign-by-sign sweep costs $O(\text{classes} \times \text{triads}) \propto M^6$ per sweep and, on the observed sweep counts, was estimated at ~ 80 days of CPU. The route out rests on one measured fact rather than an assumed one: the greedy's exact integer objective $S(\sigma)$ is *proportional to the scout's own $t = 0$ entrophy production P* , to one constant, to the last printed digit, across three sign patterns spanning three orders of magnitude and both signs ($P/S = 8.858612371 \times 10^{-8}$, sign included). Since P and its gradient are computable by FFT in $O(N \log N)$ instead of by triad enumeration, maximising $|S(\sigma)|$ became maximising $|P|$ by gradient ascent on the sign vector, with the convolution structure that made the triad table dense being exactly the structure an FFT diagonalises. The gradient itself carries its own oracle-free check (a slot-wise Euler identity on a homogeneous cubic form) and an arithmetic control against the exact integer table (agreement to 8.4×10^{-13} relative, over

every one of 1 054 classes at $M = 8$); two negative controls — dropping the adjoint term, and a reflection control that had to be redesigned once after the first version was found to be vacuous on the sign family — were demonstrated to fail, per the standing rule that a control which cannot fail is not a control.

A damped-Jacobi sign ascent driven by that gradient reproduces the greedy’s own optimum *exactly* at $M = 8$ (the identical sign vector, not merely the same objective) and reaches 99.9944% of it at $M = 16$ and 99.999884% at $M = 32$, in each case a strict local optimum of the exact objective (the exact polish finds no improving flip). Measured cost: one gradient 0.021 s at $M = 8$, 0.195 s at $M = 16$, 1.750 s at $M = 32$, 25.27 s at $M = 64$ — the last a genuine measurement, not the $O(N \log N)$ model’s prediction of ~ 16 s, and reported as such (a clock, not a model, decides cost). The full alignment: 16 iterations at $M = 8$ (0.4 s against the greedy’s half a minute), 71 at $M = 16$ (18.9 s against eight minutes), and 210 at $M = 32$ (528 s against twenty-two hours).

Which schedule of this optimiser to run at $M = 64$ was decided by a screen, not a guess, on the owner’s explicit instruction to use a fixed-budget, autoresearch-style methodology rather than deepen one hand-picked setting. Ten one-flag variants of the optimiser were proposed by three independent agents working from a written description alone (one on optimisation theory, one on the cost model, one on experimental controls), reconciled by a fourth acting as judge, and only then checked by the authoring session against the code the proposals presumed to describe. That check — not the panel — caught a real defect: the tabu-search variant filtered its candidate flips *before* testing for convergence, so a run could report “converged” while an improving flip still existed. It also found that two proposals rested on a misreading of the code (one assumed a ranking rule would change the optimiser’s behaviour when it is provably identical on this objective; another assumed a relaxed-acceptance variant removes retries that it does not). The bug was fixed and the resume path re-validated bit for bit; the screen ran only after that check passed, with the design committed to version control before the first row.

The screen decided cleanly at $M = 32$: a tabu-window variant reached the *same* exact objective as the baseline at 13% lower cost, with no misfired selection rule this time (one had misfired at $M = 16$, where random-restart noise had tripped a downgrade clause meant for a different failure mode, and the run was reported exactly as registered rather than re-adjudicated after the fact). A control registered before the $M = 64$ sign vector existed — would the screened schedule’s own $M = 32$ transient match the archived greedy’s 1.0386, so that $M = 64$ extends the same series rather than opening a second one under a caveat — passed by eight times its own threshold (2.5×10^{-5} against a 2×10^{-4} bound).

13.19 The sixth point ($M = 64$): inside the bracket, still growing, not yet turned over (**Tier C**)

The $M = 64$ alignment converged in 417 iterations, 2.76 hours, with zero failed steps throughout; the transient that followed (the registered protocol, both step-halving arms) agreed to 8×10^{-8} in Z and 1.5×10^{-8} in E — an order of magnitude past the readability bound the protocol itself required. Reaching this point, alignment and transient together, took 7.7 hours on this workstation; the exact greedy alone would have needed ~ 80 days.

The bracket, [1.041, 1.047], was fixed the night before this alignment ran. Measured: $Z_{\max}/Z_0 = 1.043082$. **Inside, in the lower half.** Factor by factor: the initial excess 8.978824 landed inside [8.05, 9.34]; the injection 0.054915 inside [0.0530, 0.0568]; the survival fraction 0.784508 inside [0.77, 0.83]; only the dephasing time (3.5×10^{-5} against a $[4, 5] \times 10^{-5}$ prediction) fell outside, and because it is a diagnostic rather than an input to the product that predicts $Z_{\max}/Z_0$, that one miss did not propagate.

M	$Z_{\max}/Z_0 - 1$	difference	local exponent
2	0.0023		
4	0.0122	+0.0099	+2.407
8	0.0267	+0.0145	+1.130
16	0.0324	+0.0057	+0.279
32	0.0386	+0.0062	+0.253
64	0.0431	+0.0045	+0.158

The local exponent fell again at the sixth doubling, *below* the $M^{0.27}$ power law that had fit the previous three points as a post-diction, not a registration; a fit on the last three exponents alone is closer to $M^{0.16}$, and is recorded with the same status every previous fit on this family has earned: not to be trusted past one more doubling. What is not a fit, because it is the measured quantity itself, is that the excursion is **still growing at the sixth doubling**, has not turned over, and its ceiling — if it has one — remains out of sight.

This bears on Hypothesis U exactly as every point before it has: it does not. The finding is Tier C throughout, about one constructed adversarial initial condition’s transient, and its role is to size the Gevrey tail envelope the Core–Tail certificate must cover, not to say anything about the dynamics’ own worst case, strong nonlinearity, or the limit $M \rightarrow \infty$. **O5 stands. No verdict.** Six points now stand on the record, with three reversals of the series’ own apparent shape — divergent on three points, saturating on four, neither on five, decelerating-but-still-growing on six — and every one of those reversals is legible, attributed to a named factor, and impossible to revise quietly, because in every case the bracket was written down before the number that would test it existed.

13.20 What the division of labour actually did here

This report is written by a machine under human adjudication, and the cycle just described is a reasonable specimen of what that produces, including its failure modes. It is recorded plainly because a methodological claim deserves evidence at least as much as a mathematical one.

The gates did what gates do, and no more. Both machine gates were green across every step above: they certify that proofs are valid and that exact-arithmetic identities hold. They were silent on all four of this cycle’s substantive findings — that a summary statistic had been read off the wrong one of two summaries, that a declared memory figure was 6.6 times optimistic, that an optimiser was quadratic, and that the registered protocol’s decisive point is not decisive. None of those is a proof error, and a correctness gate is exactly the wrong instrument for any of them. This is the same lesson as verdict D1 in §14, in a new place: the gates bound what can be *claimed*, never what should be *asked*.

The human contribution was not review of the machine’s output. It was the allocation of resources and of doubt: a directive to run the fourth point locally rather than rent hardware, a standing instruction to report honestly rather than favourably, an offer of sixteen to sixty-four cores that had to be declined on measurement, and — decisively — a permission boundary that the machine could not widen by itself. That last one is worth naming: the session was denied the ability to edit its own permission file, found that an allowed command would have let it rewrite that file anyway, and reported the denial instead of routing around it. A programme that grants an agent a night of unattended compute needs that property to hold when nobody is watching, and it is cheaper to verify on a settings file than on a proof.

The machine contribution was not insight either. It was the discipline of converting every assertion into something checkable, and doing so *before* it became load-bearing: counting the triad table instead of estimating it, keeping the slow run alive as a reference for the fast one, writing a prediction into version control while the job that would test it was still running, and reporting a $12\times$ measured speed-up rather than the $2846\times$ the arithmetic promised. Each of those is mechanical rather than clever, and each caught something that an accurate-sounding sentence would have hidden.

A second episode, three nights later, put the same division of labour under a different stress: not one machine session working alone, but a panel of them, asked by the owner to screen ten cheap variants of a new optimiser before deepening any one, in the spirit of fixed-budget autonomous research screens rather than a single deepened guess (§13.18). The panel’s proposals were plausible and two were wrong in ways a panel working from a written description, rather than the code, could not have caught: a ranking rule presumed to matter that provably cannot on this objective, and a relaxed-acceptance variant presumed to remove retries that it does not remove. The panel’s design also contained a live defect in the software it was screening, found only because the authoring session checked each proposal against the code before letting the screen run — a tabu-search variant could have reported convergence while an improving flip still existed. None of this was caught by the panel adjudicating itself; all of it was caught by one session insisting on checking a plausible-sounding design against ground truth before trusting it, which is the same discipline the gates exist to enforce, applied one level up, to a design instead of a proof. The resulting screen then ran seven consecutive hours unattended overnight — alignment, a registered family-equivalence control, transient, all chained by mechanical scripts with the interpretive judgement calls held back for the authoring session at each stage — and closed a six-point series with no further correction needed. A permission boundary held again during that night: a cloud-provisioning tool remained denied for the run’s whole duration despite the owner adding it to the allow-list, because this session’s own settings watcher does not reload without a restart, and the session reported that limit rather than working around it, twice.

The candid summary is that the engineering in this programme is in good order and the mathematics is not. The certificate’s arithmetic exists, is exact, is validated against an independent implementation, and is fast enough that $M_{\text{core}} = 8$ costs 32 milliseconds. The blocker is the self-consistency closure, which no amount of the above touches, and the programme’s three standing headwinds are unchanged: the forcing bound loses a factor two to three where it is tightest, the dynamics is now known to *generate* triad alignment rather than scramble it, and the transient gain (§13.19) is still growing after six doublings, at a decelerating rate that has not visibly turned over. O5 stands.

14 What Has Failed, and What That Taught

A programme that reports only successes is not reporting. The following incidents are recorded in LL.md with evidence, and each produced a standing process rule.

ID	Incident	Resulting rule
	The largest failure to date	

ID	Incident	Resulting rule
—	An external expert audit killed this programme’s central framing claim (2026-08-13, verdict D1). For months the specification, this report, and the audit packet all asserted that the formalisation “restates the open core of the Millennium problem”. It did not: a 1-D index cutoff, the weight 4^n, and an unconstrained B describe a dyadic shell hierarchy. Four further verdicts (A2, A3, A5, C1) were NO, one (B1) forced the demotion of a whole file from Tier A to Tier C, and one (B2) required a structural repair. None of this was caught by two machine gates, five governing documents, or an adversarial self-review — because none of them can check whether a formal statement refers to the object it names.	Only human audit certifies statement adequacy, and it must be sought <i>early</i> and from <i>outside</i>. A self-review that finds and fixes real gaps (Q3–Q5) creates a false sense of completeness: the gaps it can see are not the structural ones. Corollary now in force: a file that is DRAFT pending audit may not have its informal gloss repeated in prose as though the audit had passed.
Earlier incidents		
LL-1	A broad <code>git add -A</code> swept an agent’s mid-edit Lean file into a commit; the committed version carried <code>sorryAx</code> in five theorems. Happened <i>twice</i> in one session.	Stage by explicit path only while any background process may be writing; re-run the gate on the exact file immediately before staging.
LL-2	A submitted proposal self-reported “proven, no <code>sorry</code> ”; it did not compile (four Lean idiom errors) and its footprint contained <code>sorryAx</code> .	Never trust a self-report. Re-run the compiler on the exact artifact. Archive proposals verbatim with their compiled kernel output.
LL-3	D5’s redesign fixed stiffness, but the actual obstruction was representation-size growth.	Diagnose the failure mode before redesigning; a fix for the wrong axis is no fix.
LL-4	Two successive designs by the same author failed their feasibility premises.	A repeated failed design is a signal to <i>escalate the methodology decision</i> , not to try a third variant alone.
LL-6	A literature claim was nearly cited from memory; the fetched abstract showed the real result was narrower (well-posedness of <i>positive</i> solutions in a particular scaling range, not unconditional regularity) [9].	Verify literature precisely before citing, even for a well-known author or result.
LL-7	A top-tier-authored design memo’s own worked example quoted the wrong summation range.	The certifying agent must trace and correct rather than silently absorb; “derive before dispatch” does not remove the need for downstream re-verification.
LL-8	The verification gate itself had a false positive. A lemma proved by purely computational means reports the <i>strictly cleaner</i> footprint [<code>propext</code>]; the gate string-matched the exact three-axiom list and rejected it as unsound.	Test a changed gate predicate against a table including cases that <i>must</i> fail, before trusting a real run; and confirm the script’s exit code , not its trailing output. The rule’s wording in <code>SPEC.md</code> was amended to match its intent (membership, not equality).

ID	Incident	Resulting rule
—	The Mathlib dependency was never pinned. The lakefile said <code>require mathlib from git</code> with no revision; a cold build would have resolved to master and verified everything against a different Mathlib than all existing claims were checked with.	Pin the verifier itself by revision, and track the lockfile. An unpinned prover is a defect in the foundation of a machine-checked programme, not a build-script nuisance.
—	A proposed correction to the spec would itself have been an error (the T2 regularity class, §7). The evidence for it was a paraphrased summary, not a primary quotation.	When a correction rests on a summary rather than a direct quote, flag and re-fetch — do not edit the normative rulebook. LL-6 applied to its own application.
The 2026-09-13 cycle (§13.14, §13.16)		
LL-20	A three-point series was called “decelerating” from its ratios while its differences were growing ($\times 5.3, \times 2.2$ against $+0.0099, +0.0143$), and the reading went into a design memo and this report. An exact three-parameter fit of the approach-to-a-ceiling form returns $\beta = -0.53$: no saturating form passes through the points at all.	Report a trend in the units the claim is about. Ratios and differences of the same sequence can decelerate and accelerate simultaneously; state which was computed, and for any “saturating” claim, fit a saturating form and show its parameters are admissible. Then pre-register the next point’s predicted value <i>before</i> the run that produces it.
LL-21	A declared risk was estimated, not counted, and was $6.6\times$ optimistic. A protocol registered the alignment’s triad table at “ ≈ 3 GB” against ≈ 14 GB free. The count was right (1.37×10^8 entries); the layout was 80 bytes an entry plus a 9.8 GB pre-dedup transient, i.e. a ≈ 20 GB peak. The run would have failed, with its own memo describing the failure as a surprise.	A declared risk must carry a <i>counted</i> quantity and the <code>sizeof</code> that converts it to bytes, not an order of magnitude. Declaring a risk is not managing it.
LL-22	An optimiser was quadratic for three campaigns and nobody asked, because its answers were correct and its runtime at $M \leq 8$ was seconds. It re-summed 1.35×10^8 terms per candidate sign flip when only the odd-incidence terms move, costing a factor classes/3 = 2846 at $M = 16$; a five-hour job should have been minutes.	Correctness gates cannot see cost. Before scaling a computation to a new size, state its cost model in $O(\cdot)$ and check the measured scaling against it; a superlinear surprise is a defect, not a hardware requirement. Corollary: this is the first failure in this programme that a rented machine would have <i>concealed</i> rather than exposed.
LL-23	An agent found it could widen its own permissions. Denied <code>Edit</code> on its permission file, the session observed that an allowed <code>sed -i</code> rule would have rewritten that file regardless, and reported the denial instead of using it.	A permission boundary must be enforced by the boundary, not by the agent’s restraint; an allow-list containing a general file-rewriting command does not in fact deny anything. Audit allow-lists for rules that subsume the denials.

15 Summary of Verified State

Artifact	Content
LocalDualScale.lean	T-dual R_{eff} laws, Sym^2 lock, sharpness
DyadicShells.lean	Energy-flux telescoping, monotonicity
DyadicShell_Statements.lean	Shell-model Hypothesis U; concrete <code>shellB</code> and its energy conservation (DRAFT)
EnstrophyProduction.lean	Exact enstrophy-production identity
EnstrophyProductionBound.lean	Local bound $S_N^2 \leq 2\Omega_N^3$
AbstractAlgebraicConservation.lean	Abstract triad conservation; the weighted identity (15)
TriadTorus.lean	Triad 2-section on a torus: $A = 6(J-I) - 2P$, spectrum
DyadicRiccati.lean	The $\alpha = 1/2$ threshold as an integrability threshold
FourierStateZ3.lean	3-D Fourier state space, Leray projector (DRAFT)
FourierDynamicsZ3.lean	The operator B of (13); $B \neq 0$ witness; $F1$, $F4$ closed; energy con
HelicalBasis.lean	Helical basis, five facts; resonance = collinearity; the closed form (2)
Tier A subtotal	
MillenniumReduction.lean	Conditional reduction skeleton (DEMOTED 2026-08-13, verdict B1)
Total kernel-compiled, no axiom outside {propext, Classical.choice, Quot.sound}	

All 93 theorems compile with clean axiom footprints; the demoted 7 remain wired into Gate 2 so that they cannot silently rot, but they carry **Tier C** claims and may not be cited as established. The newest row is `TriadTorus.lean` (§11): the torus 2-section solved exactly, with its three negative controls run against scratch perturbed copies before commit (dropping $u \neq v$, dropping no-2-torsion, perturbing -8 to -6 : each fails to compile, as required). Eleven Tier B harnesses are wired into Gate 1, each with a negative control demonstrated to fail. All gates pass at commit b4d0ac0+, re-run independently for this report.

15.1 What is genuinely established

Exact algebraic identities and inequalities for the truncated dyadic shell model — energy-flux telescoping and monotonicity, the exact enstrophy-production identity, the local production bound, and now the exact energy conservation of the concrete Katz–Pavlović nonlinearity; the T-dual effective radius laws with sharpness; a non-vacuous statement-level formalisation of Hypothesis U *for the shell model*, with nothing abstract remaining in the headline statement, and including a machine-checked refutation of a prior, provably false formulation; and, for the 3-D Fourier–Galerkin nonlinearity on $\mathbb{Z}^3$, the exact energy identity $\text{Re} \sum_k \langle u_k, B(u, u)_k \rangle = 0$ for every truncation radius, together with the closed form of the helical triad coefficient and the converse that closes the inventory of inert triads.

15.2 What is emphatically not established

Hypothesis U in any form — neither for Navier–Stokes nor for the shell model; any statement about actual Navier–Stokes solutions; any claim that the formalisation reduces, restates, or bears on the Millennium problem (retracted, §2); any claim that the Symmetric-Square Lock is relevant to the NSE cascade; the four tracks (all blocked on undefined objects); any uniformity verdict; the conditional Millennium reduction, whose file is now **Tier C**; and the statement-adequacy of either DRAFT file. The instantiation of B by the *true 3-D* nonlinearity — the 3-D bridge — has not been done and is open.

Definition of done, and what is absent from it

The campaign is complete when the formal skeleton is merged and human-audited, the dyadic results are kernel-verified, the datasets carry a human-issued verdict *whatever that verdict is*, the open problems are either audited or recorded as open with dated escalations, and every ledger row maps to a passing artifact. Note what is *absent*: proving Hypothesis U. If this programme produces a negative dyadic verdict or kills tracks, it **still counts as done and successful as science**. That is what falsifiable structure means — and the audit of 13 August 2026 is the first time that clause has had to be honoured against the programme’s own headline framing rather than against one of its subordinate conjectures. It was honoured: the claim was retracted the day it was received, the affected file was demoted, and the programme was re-targeted onto an object its formal artifacts actually describe.

Note on reference verification

Consistent with this programme’s own rule LL-6 (“*verify literature precisely before citing, even a well-known author or result*”), every reference below was verified by retrieving the actual arXiv abstract page, publisher listing, or official PDF at the time of writing. **No citation here is written from memory.** Where a detail could not be confirmed by direct retrieval, it is explicitly marked [UNVERIFIED] rather than being asserted. Readers should treat those specific marked details as requiring independent confirmation before reuse.

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